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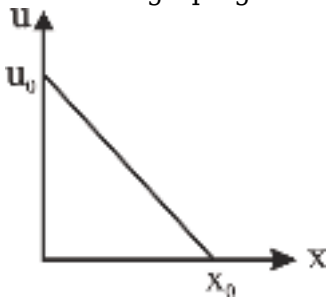
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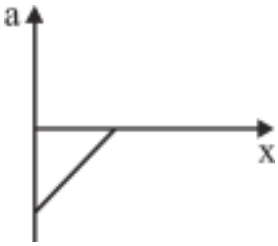
PHYSICS

1) The given graph shows the variation of velocity with displacement. Which one of the graph given

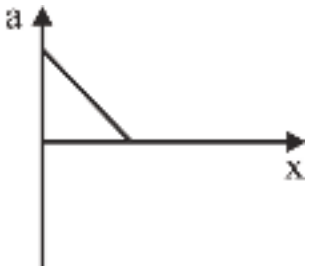


below correctly represents the variation of acceleration with displacement :-

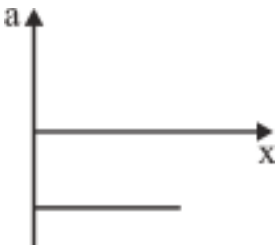
(1)



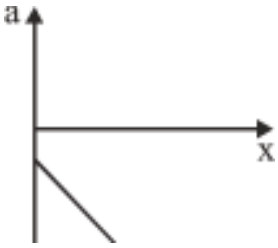
(2)



(3)



(4)



2) A stone is dropped from the 25th storey of a multi-storey building and it reaches the ground is 5s. In the first second, it passes through how many storeys of the building :- ($g = 10\ m/s^2$)

(1) 1

(2) 2

(3) 3

(4) 5

3) A body of mass m is projected horizontally with a velocity v from the top of a tower of height h and it reaches the ground at a distance x from the foot of the tower. If a second body of mass 2m is projected horizontally from the top of a tower of height 2h, it reaches the ground at a distance 2x from the foot of the tower. The horizontal velocity of the second body is :-

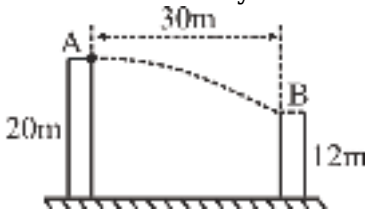
(1) v

(2) 2

(3) $\sqrt{2}$ v

(4) $\frac{v}{2}$

4) A boy wants to throw a ball from a point A so as to just reach point B as shown. The minimum horizontal velocity with which the boy should throw the ball is [g = 10 ms^{-2}]



(1) 2.4 ms^{-1}

(2) 23.7 ms^{-1}

(3) 238 ms^{-1}

(4) 47.4 ms^{-1}

5) A person in a stationary vehicle on a road has to hold his umbrella at 60° with the vertical to keep the rain away. He throws the umbrella and drives the vehicle at 20 ms^{-1}. He finds the rain drops are falling vertically on him. The speed of the rain drops with respect to the road is :

(1) $\frac{40}{\sqrt{3}}\ ms^{-1}$

(2) $\frac{20}{\sqrt{3}}\ ms^{-1}$

(3) $\frac{30}{\sqrt{3}}\ ms^{-1}$

(4) $\frac{25}{\sqrt{3}}\ ms^{-1}$

6)

The position of a particle moving along x-axis is given as
$x = 4(t-2) + a(t-2)^2$
Which of the following is true ?

(1) The initial velocity of particle is 4

(2) The acceleration of particle is 2a

(3) The particle is at origin at t = 0

(4) None of these

7) The position x of a particle with respect to time t along x-axis is given by: $x = 9t^2 - t^3$, where x is in metres and t in second. What will be the position of this particle when it achieves maximum speed along the +x direction?

(1) 54 m

(2) 81 m

(3) 24 m

(4) 32 m

8) The relation between time and distance is $t = \alpha x^2 + \beta x$, where α and β are constants. The retardation is :-

(1) $2\alpha v^3$

(2) $2\beta v^3$

(3) $2\alpha\beta v^3$

(4) $2\beta^2 v^3$

9) The velocity of an object moving rectilinearly is given as a function of time by $v = 4t - 3t^2$, where v is in m/s and t is in seconds. The average velocity of particle between t = 0 to t = 2 seconds is :-

(1) 0

(2) –2m/s

(3) –4m/s

(4) None of these

10) Velocity vector of a particle is given as $\vec{V} = 4t\hat{i} + 3\hat{j}$. At t = 0 position of the particle is given as $\vec{r}_0 = 2\hat{j}$. Find the position vector of the particle at t = 2 sec. :-

(1) $8\hat{i} + 6\hat{j}$

(2) $8\hat{i} + 8\hat{j}$

(3) $8\hat{i} + 4\hat{j}$

(4) None

11) A body is placed on a rough inclined plane of inclination θ. As the angle θ is increased from 0° to 90°, the contact force between the body and the plane:-

(1) remains constant

(2) remains constant first and then decreases finally to zero

(3) decreases first and then increases

(4) increases first and then decreases

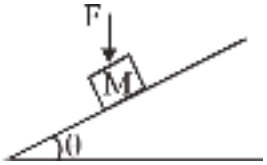
12) **Assertion :-** In a tug-of war game, the two teams pull a massless rope with same magnitude of force
Reason :- The force exerted by any team on massless rope is equal to tension developed in the rope which will be same at both ends

(1) Both assertion and reason are true and the reason is the correct explanation of assertion.
(2) Both assertion and reason are true and the reason is not the correct explanation of assertion.
(3) Assertion is true but reason is false.
(4) Assertion is false but reason is true.

13) A force $\vec{F}_1$ acts on a particle so as to accelerate it from rest to a velocity υ. The force $\vec{F}_1$ is then replaced by $\vec{F}_2$ which decelerates it to rest :-

(1) $\vec{F}_1$ must be unequal to $\vec{F}_2$
(2) $\vec{F}_1$ may be equal to $\vec{F}_2$
(3) $\vec{F}_1$ must be equal to $\vec{F}_2$
(4) None of these

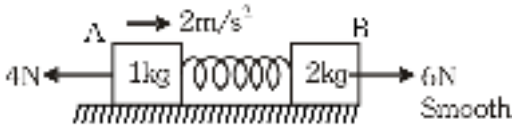
14) A block of mass M is sliding down the plane. Coefficient of static friction is μ_s and kinetic friction is μ_k. Then friction force acting on the block is :-



(1) (F + Mg) sin θ
(2) μ_k(F + Mg) cos θ
(3) μ_s mg cos θ
(4) (Mg + F) tan θ

15)

In shown system acceleration of block 'B' :



(1) 6 m/s^2
(2) 2 m/s^2
(3) 1 m/s^2
(4) 0

16) The rate of mass of the gas emitted from rear of a rocket is initially 0.1 kg/sec. If the speed of the gas relative to the rocket is 50 m/sec and mass of the rocket is 2 kg, then the acceleration of the rocket (in m/sec^2) is :- (Rocket is in gravity free space)

(1) 5
(2) 5.2
(3) 2.5
(4) 25

17) A force is applied to two different objects. It produces accelerations of 3 m/s^2 and 6 m/s^2 respectively in these objects. If the same force is applied to the combination of these objects then acceleration of combination will be :-

(1) 2 m/s^2

(2) 4.5 m/s^2

(3) 4 m/s^2

(4) none

18) Which of the following forces can be self adjusting :

(1) Static friction

(2) Normal reaction

(3) Tension in string

(4) All of the above

19) A ball of mass m, moving with velocity $\left(2\hat{i}+3\hat{j}\right)$ m/s recieves an impulse $\left(-3m\hat{i}\right)$ N-s due to an impact. What is the velocity of the ball just after the impact ?

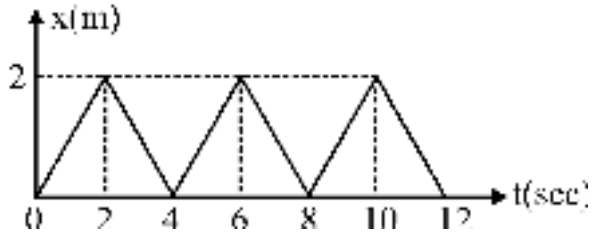
(1) $\left(-\hat{i}+3\hat{j}\right)$ m/s

(2) $\left(\hat{i}-3\hat{j}\right)$ m/s

(3) $\left(-\hat{i}-3\hat{j}\right)$ m/s

(4) $\left(\hat{i}+3\hat{j}\right)$ m/s

20) Figure shows a position time graph for a body of mass 0.2 kg in one dimensional motion. The magnitude of each impulse will be :-



(1) 0.2 Ns

(2) 0.4 Ns

(3) 0.8 Ns

(4) 1.6 Ns

21) A ball of mass m is moving towards a batsman at a speed v. The batsman strikes the ball and deflectes it by an angle θ without changing its speed. The impulse imparted to the ball is given by :-

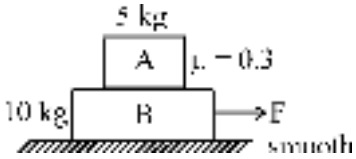
(1) mv cosθ

(2) mv sinθ

(3) $mv\cos\frac{\theta}{2}$

(4) $2mv\sin\frac{\theta}{2}$

22) The maximum value of F so that there is no slipping between A and B is



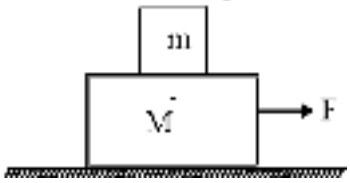
(1) 35 N

(2) 15 N

(3) 100 N

(4) 45 N

23) A smooth block of mass m is placed on a smooth plank of mass M. When a force F is applied on the plank its acceleration will be :-



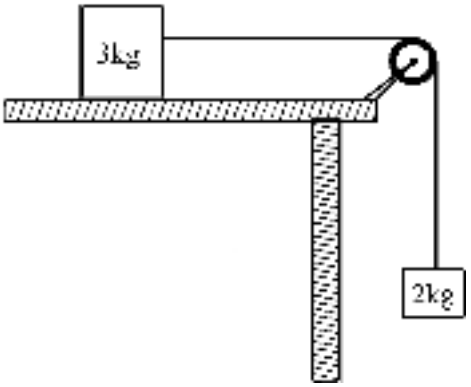
(1) $\frac{F}{M+m}$

(2) $\frac{F}{m}$

(3) $\frac{F}{M}$

(4) $\frac{MF}{(M+m)^2}$

24) Consider a body of mass 3kg at rest on a smooth horizontal table. This body is connected by a light string, which passes over a smooth pulley at the edge of the table, to another body of mass 2 kg hanging freely. This 2 kg mass is released from rest. Now consider the following statements :-



(A) The masses remains at rest

(B) The 3 kg mass moves uniformly while 2 kg mass moves with acceleration $\frac{2}{5}$g m/s^2

(C) Both bodies move with acceleration $\frac{2}{5}$g m/s^2

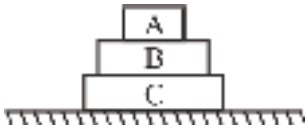
(D) The tension in the string near the first body is more than that near the second body

(E) The tension in the string is $\frac{6g}{5}$ N

Then the correct statements are

(1) B and D only
(2) C and E only
(3) B, D and E only
(4) A only

25) Three blocks are placed as shown in figure below. The mass of A,B and C are m_1,m_2 and m_3, respectively. The force exerted by block C on B is:-



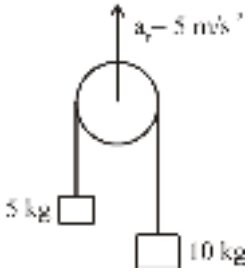
(1) m_1g
(2) $(m_1 + m_2)g$
(3) m_2g
(4) $(m_1 + m_2 + m_3)g$

26) A block of mass m is suspended from a fixed support with the help of a cord. Another identical cord is attached to the bottom of the block. Which of the following statement is true?



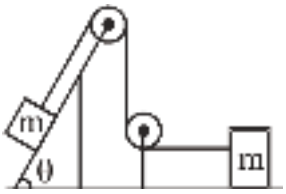
(1) If the lower cord is pulled suddenly, only the upper cord will break.
(2) If the lower cord is pulled suddenly, only the lower cord will break.
(3) If pull on the lower cord is increased gradually, only the lower cord will break.
(4) None of these

27) Two blocks of masses 5 kg and 10 kg are attached to a string passing over pulley which is accelerated upwards by 5 m/s^2 then acceleration of blocks with respect to pulley will be :



(1) $\frac{10}{3}$ m/s^2
(2) 5 m/s^2
(3) 10 m/s^2
(4) 0 m/s^2

28) For the system shown in figure, the pulleys are light and frictionless. The tension in the string

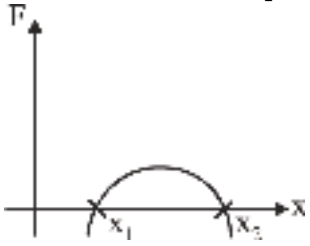
will be :- 

(1) $\frac{2}{3}$ mg sinθ

(2) $\frac{3}{2}$ mg sinθ

(3) 2 mg sinθ

(4) $\frac{1}{2}$mg sinθ

29) The force acting on a body moving along x-axis varies with the position of the particle as shown in figure. The body is in stable equilibrium at :- 

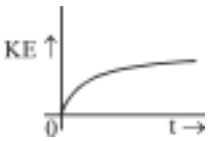
(1) $x = x_1$

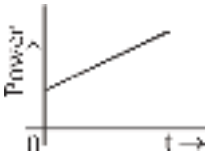
(2) $x = x_2$

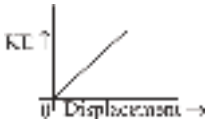
(3) Both $x = x_1$ and $x = x_2$

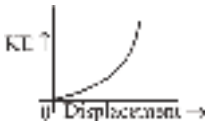
(4) Neither $x = x_1$ nor at $x = x_2$

30) A block is resting over a smooth horizontal plane. A constant horizontal force starts acting on it at t=0. Which of the following graph is correct :-

(1) 

(2) 

(3) 

(4) 

31) An elastic string of unstretched length L and force constant k is stretched by a small length x. It is further stretch by small length y. The work done in the second stretching is :-

(1) $\left(\frac{1}{2}\right) ky^2$

(2) $\left(\frac{1}{2}\right) k\left(x^2 + y^2\right)$

(3) $\left(\frac{1}{2}\right) k(x + y)^2$

(4) $\left(\frac{1}{2}\right) ky\,(2x + y)$

32) If the force applied is F and the velocity gained is $\frac{V}{2}$, then the average power developed will be :

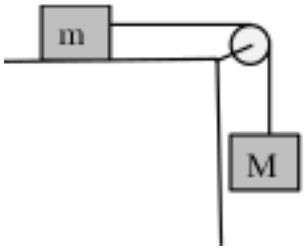
(1) $\frac{F}{V}$

(2) $\frac{V}{F}$

(3) $\frac{FV}{2}$

(4) $\frac{FV}{4}$

33) A mass m, lying on a horizontal frictionless surface is connected to a mass M as shown in figure. The system is now released. The velocity of mass m when mass M descends a distance h is



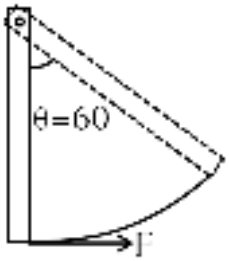
(1) $\sqrt{\frac{2Mgh}{m}}$

(2) $\sqrt{\frac{2mgh}{M}}$

(3) $\sqrt{\frac{2Mgh}{(M + m)}}$

(4) $\sqrt{2gh}$

34) Find the work done by applied force F by slowly bringing the rod to the inclined position. mass of rod is 5 kg & length 1 m. ($g = 10$ m/sec^2) :-



(1) 205 J

(2) 5 J

(3) 10 J

(4) 12.5 J

35) If a simple pendulum of length L has maximum angular displacement α, then the maximum kinetic energy of bob of mass M is :-

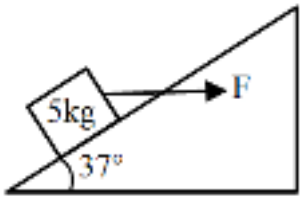
(1) $\frac{mL}{2g}$

(2) $\frac{mg}{2L}$

(3) mgL(1–cos α)

(4) mgL sin (α/2)

36) A block of mass 5 kg is pulled up an inclined plane with constant velocity by applying a force as shown in figure. If the block gets displaced by 2 m, the work done by force F is :-



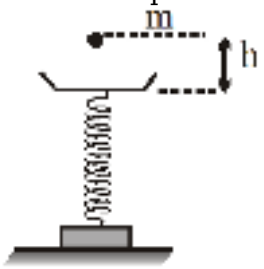
(1) 60 J

(2) 30 J

(3) 90 J

(4) 15 J

37) A ball of mass m is dropped from a height h on a platform fixed at the top of a vertical spring.



The platform is displaced by a distance x. The spring constant is

(1) $\frac{2mg}{x}$

(2) $\frac{2mgh}{x^2}$

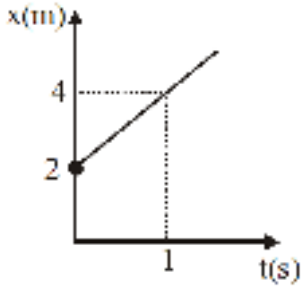
(3) $\frac{2mg(h+x)}{x^2}$

(4) $\frac{2mg(h+x)}{h^2}$

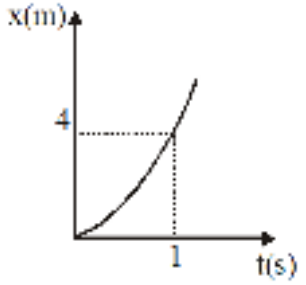
38) A car is moving on a test track which is x-axis. At the beginning the car is at x = 2. The velocity v of the car as function of position the car x, is v = 2x m/s, where x is in meters. The correct

representation of the motion is :-

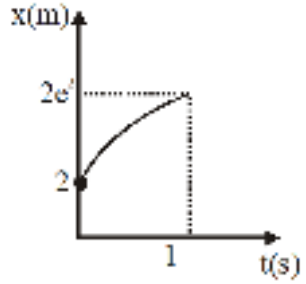
(1)



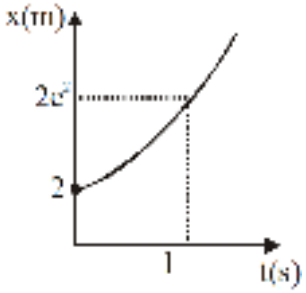
(2)



(3)



(4)



39) The potential energy of a particle of mass 5 kg moving in the x – y plane is given by U = (–7x+ 24y) J, where x and y are given in metre. If the particle starts from rest, from the origin, then the speed of the particle at t=2s is

(1) 5 m/s

(2) 14 m/s

(3) 17.5 m/s

(4) 10 m/s

40) A body is displaced from (0, 0) to (1m, 1m) along the path x = y by a force $\vec{F} = \left(x^2\hat{j} + y\hat{i}\right) N$. The work done by this be :

(1) 4/3 J

(2) 5/6 J

(3) 3/2 J

(4) 7/5 J

41) A uniform chain of length ℓ and mass m overhangs from a smooth table so that $\frac{2}{3}$rd part of it is on the table then velocity of chain when it completely slips off the table :-

(1) $\sqrt{2g\ell}$

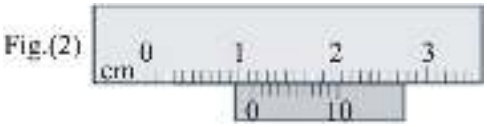
(2) $\frac{2}{3}\sqrt{2g\ell}$

(3) $\sqrt{\frac{2}{3}g\ell}$

(4) None

42) The main scale of a Vernier calliper reads in mm and its Vernier is divided into 10 divisions which coincide with 9 divisions of the main scale. The reading for the situation is found to be (x/10) mm. Find the value of x. In fig (1) 8th div. of V.S. coincides with M.S.D. & in fig (2) 6th div. of V.S. coincides with M.S.D.





(1) 104

(2) 108

(3) 114

(4) 118

43) The pitch of a screw gauge is 0.5 mm and there are 100 divisions on its circular scale. The instrument reads +2 divisions on circular scale when nothing is put in-between its jaws. In measuring the diameter of a wire, there are 8 divisions on the main scale and 83rd division of circular scale coincides with the reference line. Then the diameter of the wire is

(1) 4.05 mm

(2) 4.405 mm

(3) 3.05 mm

(4) 1.25 mm

44) A screw gauge has 50 divisions on its circular scale. The circular scale is 4 units ahead of the pitch scale marking, prior to use. Upon one complete rotation of the circular scale, a displacement of 0.5 mm is noticed on the pitch scale. The nature of zero error involved, and the least count of the screw gauge, are respectively :

(1) Negative, 2 μm

(2) Positive, 10 μm

(3) Positive, 0.1 μm

(4) Positive, 0.1 mm

45) A student measured the length of a rod and wrote it as 3.50 *cm*. Which instrument did he use to measure it?

(1) A screw gauge having 100 divisions in the circular scale and pitch as 1 *mm*.

(2) A screw gauge having 50 divisions in the circular scale and pitch as 1 *mm*.

(3) A meter scale.

(4) A vernier calliper where the 10 divisions in vernier scale matches with 9 division in main scale and main scale has 10 divisions in 1 *cm*.

CHEMISTRY

1) The equivalent weight of phosphoric acid (H_3PO_4) in the reaction,
$NaOH + H_3PO_4 \rightarrow NaH_2PO_4 + H_2O$ is :-

(1) 59

(2) 49

(3) 25

(4) 98

2) When 1L of CO_2 is heated with graphite, the volume of the gases collected is 1.5 L. Calculate the number of moles of CO produced at S.T.P.:-

(1) $\frac{1}{11.2}$

(2) $\frac{28}{22.4}$

(3) $\frac{1}{22.4}$

(4) $\frac{14}{22.4}$

3) If $Ca_3(PO_4)_2$ and H_3PO_3 samples contain same number of 'P' atoms then the ratio of oxygen atoms in these compounds respectively is?

(1) 8/3

(2) 2/3

(3) 3

(4) 4/3

4) The conversion of oxygen to ozone occurs to the extent of 15% only. The mass of ozone that can be prepared from 67.2 L of oxygen at 1 atm and 273 K will be :-

(1) 14.4 gm

(2) 96 gm

(3) 640 gm

(4) 64 gm

5) A metal oxide has the formula M_2O_3. Its 0.1595 g requires 6 mg of hydrogen for complete reduction into free metal and water. What is the atomic mass of metal ?

(1) 54.4

(2) 46.56

(3) 55.8

(4) 58.5

6) A gaseous mixture of propane and butane of volume 3 litre on complete combustion produces 11.0 litre CO_2 under standard conditions of temperature and pressure. The ratio of volume of butane to propane is :-

(1) 1:2

(2) 2:1

(3) 3:2

(4) 3:1

7) The equivalent weight of a metal is 9 and vapour density of its chloride is 59.25 then Atomic weight of metal would be :-

(1) 23.9

(2) 27.3

(3) 36.3

(4) 48.3

8) The crystalline salt $Na_2SO_4.xH_2O$ on heating loses 50.35% of its mass and become anhydrous. The formula of crystalline salt is.

(1) $Na_2SO_4.10H_2O$

(2) $Na_2SO_4.8H_2O$

(3) $Na_2SO_4.5H_2O$

(4) $Na_2SO_4.H_2O$

9) The iron (III) oxide, Fe_2O_3 reacts with CO to give iron and CO_2
$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$
The reaction of 480.0 g Fe_2O_3 with excess of CO yields 252.0 g iron. Thus, percentage yield is.
(Fe = 56)

(1) 100.0%

(2) 91.2%

(3) 75.0%

(4) 50.0%

10) Hemoglobin contains 0.34% of iron by mass. The number of Fe atoms in 3.3 g of hemoglobin is :
(Given : Atomic mass of Fe is 56 u, N_A in 6.022×10^{23} mol^{-1})

(1) 1.21×10^5

(2) 12.0×10^{16}

(3) 1.21×10^{20}

(4) 3.4×10^{22}

11)

For the reaction
$SnO_2(s) + 2H_2(g) \rightleftharpoons 2H_2O(g) + Sn(l)$

calculate K_p at 900 K, where the equilibrium steam-hydrogen mixture contains 45% H_2 by volume:-

(1) 1.49

(2) 1.22

(3) 0.67

(4) None of these

12)

At 1050 Kelvin, For chemical reaction
$FeO(s) + CO(g) \rightleftharpoons Fe(s) + CO_2(g)$; $K_P = 0.25$
What are the equilibrium partial pressure of CO(g) and CO_2(g) at 1050 Kelvin, if the initial partial pressure are ;
$P_{CO(g)}$ = 1.6 atm and $P_{CO_2(g)}$ = 0.8 atm :-

(1) 0.52 atm, 0.95 atm

(2) 0.86 atm, 1.16 atm

(3) 2.12 atm, 0.38

(4) 1.92 atm, 0.48 atm

13)

If K_1, K_2, K_3 are equilibrium constant for formation of AD, AD_2, AD_3 respectively as follows
$A+D \rightleftharpoons AD$, $AD+D \rightleftharpoons AD_2$,
$AD_2 + D \rightleftharpoons AD_3$. Then equilibrium constant 'K' for $A + 3D \rightleftharpoons AD_3$ is related as

(1) $K_1 + K_2 + K_3 = K$

(2) $\log K_1 + \log K_2 + \log K_3 = \log K$

(3) $K_1 + K_2 = K_3 + K$

(4) $\log K_1 + \log K_2 = \log K_3 + \log K$

14) Degree of dissociation & equilibrium pressure for given reaction $N_2O_3 \rightleftharpoons NO + NO_2$ is $\sqrt{0.2}$ & 10 atm respectively then calculate K_p.

(1) 50 atm

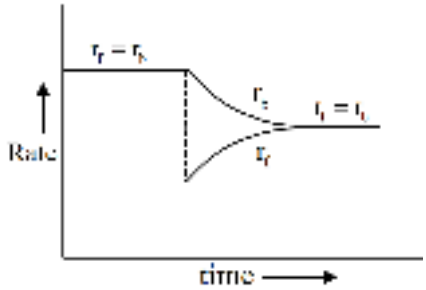
(2) 2 atm

(3) 2.5 atm

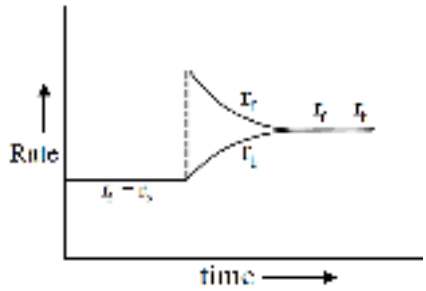
(4) 25 atm

15) To disturb the following equilibrium $A_{(g)} + B_{(g)} \rightleftharpoons C_{(g)} + D_{(g)}$ some amount of A(g) added at equilibrium and after some time system established new equilibrium. Select the graph which best represent above situation ?

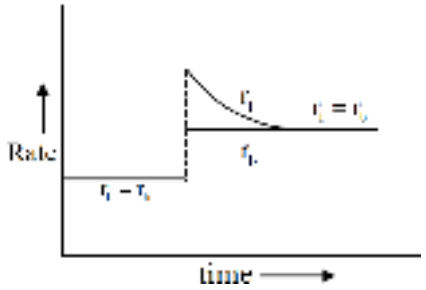
(1)



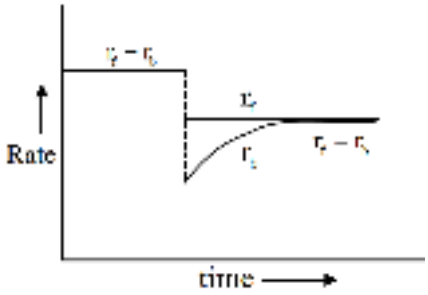
(2)



(3)



(4)



16) If the amount of dissociation is $\sqrt{0.5}$, the value of K_P for the reaction $N_2O_{3(g)} \rightleftharpoons NO_{(g)} + NO_{2(g)}$ will be

(1) Equal to pressure of the system

(2) $\frac{2}{3}$ of the pressure of the system

(3) $\frac{8}{3}$ of the pressure of the system

(4) 5 times of the pressure of the system

17) 0.2M NaOH is added to 50 mL of 0.2M acetic acid ($K_a = 1.8 \times 10^{-5}$). The volume of 0.2M NaOH required to make the pH of solution 4.74, is :-

(1) 11.5 mL

(2) 25 mL

(3) 37.5 mL

(4) 65 mL

18) Solubility of BaF_2 in a solution of $Ba(NO_3)_2$ will be represented by the concentration term :–

(1) $[Ba^{+2}]$

(2) $[F^-]$

(3) $\frac{1}{2}[F^-]$

(4) $2[NO_3^-]$

19)

Given : Enthalpy of ionization of two acids :

$\Delta H°$ (HCN) = 45.2 kJ mol^{-1}

$\Delta H°$ (CH_3COOH) = 2.1 kJ mol^{-1}

Which relationship for the two acids is true

(1) pK_a (HCN) = pK_a (CH_3COOH)

(2) pK_a (HCN) > pK_a (CH_3COOH)

(3) pK_a (HCN) < pK_a (CH_3COOH)

(4) $pK_a(HCN) = \frac{45.2}{2.1} pK_a(CH_3COOH)$

20) K_{sp} of $Mg(OH)_2$ is 1.8×10^{-11} then calculate solublity of $Mg(OH)_2$ in 0.1 M $Ba(OH)_2$ is :-

(1) 1.8×10^{-9}

(2) 0.9×10^{-11}

(3) 0.45×10^{-11}

(4) 0.45×10^{-9}

21) What is the concentration of Pb^{2+} when $PbSO_4$ ($K_{sp} = 1.8 \times 10^{-8}$) begins to precipitate from a solution that is 0.0045 M in SO_4^{2-} ?

(1) 4.0×10^{-8} M

(2) 1.0×10^{-6} M

(3) 2.0×10^{-8} M

(4) 4.0×10^{-6} M

22) Which of the following mixtures will act as buffer ?

(1) HCl + NaOH (1.5 : 3 molar ratio)

(2) HCN + NaOH (1.0 : 2 molar ratio)

(3) NH_4OH + HCl (4 : 1 molar ratio)

(4) NH_4OH + HCl (4 : 5 molar ratio)

23) CH_3NH_2 (0.12 mole, $pK_b = 3.3$) is added to 0.08 moles of HCl and the solution is diluted to one litre. At 25°C resulting pH of solution is :

(1) 10.7

(2) 3.6

(3) 10.4

(4) 11.3

24) A buffer solution is prepared by mixing 10 ml of 1.0 M acetic acid and 20ml of 0.5 M sodium acetate and then diluted to by 100 ml by adding distilled water. If K_a of acetic acid is 1.8×10^{-5} then what will be the pH of final solution ?

(1) 4.34

(2) 4.74

(3) 5.21

(4) 5.86

25) What amount of sodium propanoate should be added to one litre of an aqueous solution containing 0.02 mole of propanoic acid ($K_a = 1.34 \times 10^{-5}$ at 25°C) to obtain a buffer solution of pH 4.75 :-

(1) 4.52×10^{-2} M

(2) 3.52×10^{-2} M

(3) 2.52×10^{-2} M

(4) 1.52×10^{-2} M

26) Silver iodide is used in cloud seeding to produce artificial rain, $AgI_{(s)} \rightleftharpoons Ag^+ (aq) + I^- (aq)$; $Ksp = 8.5 \times 10^{-7}$
$AgNO_3$ and KI are mixed to give $[Ag^+] = 0.010M$;
$[I^-] = 0.015$ M. Will AgI precipitate?

(1) Yes

(2) No

(3) Can't say

(4) Depends on $[NO_3^-]$ and $[K^+]$

27)

A 0.1 M solution of ammonium acetate is diluted by adding 100 ml of water. The pH of the resulting solution will be (pK_a of acetic acid is equal to pK_a of NH_4OH)

(1) 5.0

(2) 4.9

(3) 10.0

(4) 7.0

28) Equivalent weight of H_3PO_2 when it disproportionate into PH_3 and H_3PO_3 is :

(1) M

(2) M/2

(3) M/4

(4) 3M/4

29) For redox reaction
a OH^- + b $Cr(OH)_3$ + c IO_3^- → d H_2O + e CrO_4^{2-} + f I^-
What are coefficient a, b and c respectively ?

(1) 4, 2, 1

(2) 2, 4, 1

(3) 3, 2, 2

(4) 1, 2, 4

30) A 2.5 mol of hydrazine (N_2H_4) loses 25 mole of electrons is being converted to a new compound X. Assuming that all of the nitrogen appears in the new compound, what is the oxidation state of nitrogen in compound X?

(1) –1

(2) –2

(3) +3

(4) +4

31) One mole of $KMnO_4$ is used for complete oxidation of $FeSO_4$, FeC_2O_4 and $H_2C_2O_4$ respectively and separately in acidic medium. Pick up the correct statement :-

(1) 5 mole $FeSO_4$ can be oxidised

(2) 5/3 mole FeC_2O_4 can be oxidised

(3) 2.5 mole of $H_2C_2O_4$ can be oxidised

(4) All

32) 6×10^{-3} mole of $KMnO_4$ reacts completely with 9×10^{-3} mole of X^{n+} in acidic medium to give XO_3^- and Mn^{2+}. The value of n will be

(1) 7/3

(2) 10/3

(3) 2/3

(4) 5/3

33) x moles of $KMnO_4$ react with y moles of $C_2O_4^{-2}$ in acidic medium the correct relation is:-

(1) $\frac{x}{y} = 1$

(2) $\frac{x}{y} = 0.4$

(3) $\frac{x}{y} = 2.5$

(4) x = 2y

34) The enthalpy of neutralisation of four acids HA, HB, HC and HD with NaOH are –13, –12, –11, –10 kcal/mol. Which salt has maximum degree of hydrolysis ?

(1) 1M NaA

(2) 1M NaB

(3) 1M NaC

(4) 1M NaD

35) Calculate the heat produced (in kJ) when 224 gm of CaO is completely converted to $CaCO_3$ by reaction with CO_2 at 27°C in a container of fixed volume.

Given : ΔH_f° ($CaCO_3$,s) = – 1207 kJ/mol ;

ΔH_f° (CaO, s) = – 635 kJ/mol ;

ΔH_f° (CO_2,g) = – 394 kJ/mol ;

[Use R = 8.3 J K^{-1} mol^{-1}]

(1) 702.04 kJ

(2) 821.96 kJ

(3) 1012 kJ

(4) 921 kJ

36) Find heat change in the reaction

$NH_{3\,(g)} + HCl_{(g)} \rightarrow NH_4Cl_{(s)}$

from the following data :

$NH_{3\,(g)}$ + aq. → $NH_{3\,(aq.)}$ ΔH = –8.4 Kcal

HCl(g) + aq. → $HCl_{(aq.)}$ ΔH = –17.3 Kcal

$NH_{3(aq.)} + HCl_{(aq.)} \rightarrow NH_4Cl_{(aq.)}$ ΔH = –12.5 Kcal

$NH_4Cl_{(s)}$ + aq. → $NH_4Cl_{(aq.)}$ ΔH = + 3.9 Kcal

(1) –42.1 Kcal

(2) –34.3 Kcal

(3) +34.3 Kcal

(4) +42.1 Kcal

37) Consider the following reactions :

(a) $H^+_{(aq)} + OH^-_{(aq)} \rightarrow H_2O_{(\square)}$, ΔH = $-X_1$ kJ mol^{-1}

(b) $H_{2(g)} + 1/2\ O_{2(g)} \rightarrow H_2O_{(\square)}$, ΔH = $-X_2$ kJ mol^{-1}

(c) $CO_{2(g)} + H_{2(g)} \rightarrow CO_{(g)} + H_2O_{(\square)}$ – X_3 kJ mol^{-1}

(d) $C_2H_{2(g)} + O_{2(g)} \rightarrow 2CO_{2(g)} + H_2O_{(\square)} + X_4$ kJ mol^{-1}

Enthalpy of formation of $H_2O_{(\square)}$ is :

(1) $+X_1$ kJ mol^{-1}

(2) $-X_2$ kJ mol^{-1}

(3) $+X_3$ kJ mol^{-1}

(4) $-X_4$ kJ mol^{-1}

38) Heat of formation of ethane –20.2 kCal mol^{-1} and heat of atomisation of C and H_2 are 179.2 kCal mol^{-1} and 52.1 kCal mol^{-1} respectively. If bond energy of C–H bond is 73.3 kCal mol^{-1}. Then caculate approx. B.E. of C–C :-

(1) 95 kCal mol^{-1}

(2) 230 kCal mol^{-1}

(3) 88 kCal mol^{-1}

(4) 540 kCal mol^{-1}

39)

The enthalpy change for conversion of 1 mol of ice at –5°C to water at 5°C will be :

(1) Negative

(2) Positive

(3) Zero

(4) None of these

40) Calculate the enthalpy change for the process :
$CCl_4(g) \rightarrow C_{(g)} + 4Cl(g)$
Given – $\Delta_{vap}H^o[CCl_4(\square)] = 30.5$ kJ mol^{-1}
$\Delta_fH^o(CCl_4(\square)) = -135.5$ kJ mol^{-1}
$\Delta_aH^o(C) = 715.0$ kJ mol^{-1}, $\Delta_aH^o(Cl_2) = 242$ kJ mol^{-1}
where, Δ_aH^o is enthalpy of atomisation

(1) 596 kJ mol^{-1}

(2) 326 kJ mol^{-1}

(3) 415 kJ mol^{-1}

(4) 1304 kJ mol^{-1}

41) If ΔH for the reaction
$2CO(g) + O_2(g) \rightarrow 2CO_2(g)$ is –543 kJ carried out in 1 L closed rigid vessel. If pressure in container changes from 70 atm to 40 atm then find ΔU for the reaction :-

(1) –557 kJ

(2) –540 kJ

(3) –546 kJ

(4) –573 kJ

42) If $\Delta H_{system} = 2000 \frac{KJ}{mol}$. Then calculate ΔS_{surr} at 400 K :-
(assume system - surrounding is at equilibrium)

(1) $-35 \frac{KJ}{mol\ K}$
(2) $-5 \frac{KJ}{mol\ K}$
(3) $-16000 \frac{KJ}{mol\ K}$
(4) $-20 \frac{kJ}{mol\ K}$

43) For the complete combustion of Methane,
$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ in bomb calorimeter the amount of heat released is 150 kJ/mol at 127°C. Assuming ideality, the enthalpy of combustion would be :

(1) 150 kJ
(2) –150 kJ
(3) +147.5 kJ
(4) –152.5 kJ

44) For the reaction takes place at certain temperature $NH_2COONH_4(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$. If equilibrium pressure is 3X bar then $\Delta_r G^0$ would be

(1) –RT ln 9 – 3RT ln X
(2) RT ln 4 – 3RT ln X
(3) – 3RT ln X
(4) – RT ln 4 – 3RT ln X

45)

One litre of an ideal gas at pressure of 5 atm and temperature of 27°C expands into a vaccum until its total volume becomes 5 litres. Which of the following is not correct for this process if this process is isothermal.

(1) Work is zero
(2) Heat transfer is zero
(3) Change in internal energy is zero
(4) Entropy change is zero

BIOLOGY

1) Select the incorrect statement.

(1) The earliest classification of plants was based on gross superficial morphological characters only

(2) The artificial classification gave equal weightage to vegetative & sexual characteristics

(3) Bentham and Hooker introduced a classification system for flowering plants

(4) In numerical taxonomy more weightage is given to the vegetative characters

2) Family has a group of related genera with still less number of similarities as compared to :-

(1) Kingdom and Division

(2) Class and Order

(3) Genus and Species

(4) Order and Genus

3) Which of the following statements about classification are true ?
(i) Members of a family have more common characters as compare to members of a genus.
(ii) Family contain less number of members as compare to order.
(iii) Species of a genus have more common characterstics as compare to species of another genera.
(iv) Division is a taxon which include related kingdoms.

(1) i, ii and iii only

(2) ii, iii and iv only

(3) i, ii, iii, iv

(4) ii & iii only

4) Read the following statement A and B and choose the correct option :
Statement A : The number of common characteristics goes on increasing from species to kingdom
Statement B : Higher the category, greater is the difficulty of determining the relationship to other taxa of same level.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both statements A and B are correct

(4) Both statements A and B are incorrect

5) Match the column :

	Column-I		**Column-II**
a	Phylogenetic system	1	Chemical constituents of plants
b	Numerical taxonomy	2	Based on all observable characters
c	Cytotaxonomy	3	Based on evolutionary relationship
d	Chemotaxonomy	4	Based on chromosome number, structure and behaviour

	a	b	c	d
(1)	3	4	1	2
(2)	3	2	4	1

(3)	2	4	1	3
(4)	3	4	2	1

(1) 1

(2) 2

(3) 3

(4) 4

6) In relation to scientific nomenclature of a species, which one is not correct ?

(1) Scientific nomenclature are in Latin language and Written in italics

(2) Genus name start with capital letter and specific epithet with small latter.

(3) Name of scientist should be seperately underlined when written with free hand.

(4) Biological names when hand written both the words should be underlined seperately.

7) Cat and Dog are present in the same class but not in the same family. They belong to same :-

(1) Species

(2) Genus

(3) Variety

(4) Order

8) Read the following statements :-
(i) Classification is a single step process.
(ii) Classification involves heirarchy of steps in which each step represent a rank or category.
(iii) Species is lowest rank.
How many of the above statements are correct?

(1) One

(2) Two

(3) Three

(4) Zero

9) In five kingdom classification, which of the following groups of organisms is placed in the same kingdom?

(1) *Chlorella*, *Amoeba*, *Paramoecium*, *Spirogyra*

(2) *Chlamydomonas*, *Amoeba*, *Gonyaulax*, *Chlorella*

(3) *Spirogyra*, *Nostoc*, *Anabaena*, *Mycoplasma*

(4) *Spirogyra*, *Chlamydomonas*, *Euglena*, *Nostoc*

10) which of the following kingdoms of five kingdom classification system have heterotrophic organisms only?

(1) Monera and Animalia

(2) Protista and Fungi

(3) Monera and Fungi

(4) Fungi and Animalia

11) The cell envelop in bacteria consists of :-

(1) Loosely bound three layered structures

(2) Tightly bound three layered structures

(3) Tightly bound two layered structures

(4) Tightly bound one layered structure

12) Consider the following statements (A–D) and choose the option, with correct ones only :-
(A) Monera as a group, shows most extensive metabolic diversity.
(B) Prokaryotes can fix atmospheric nitrogen as symbionts or free-livings.
(C) Bacteria can not survive in harsh habitats
(D) Bacteria can be photoautotrophic and chemoautotrophic

(1) A,B,D

(2) A,C,D

(3) C,D only

(4) only C

13) Select the **wrong** statement :-

(1) Bacterial cell wall is made up of peptidoglycan

(2) Pili and fimbriae are mainly involved in motility of bacterial cells

(3) Cyanobacteria lack flagellated cells

(4) Mycoplasma is a wall-less micro-organism

14) Which plant used as a packing material for sending seeding, flowers and living plants :-

(1) *Sphagnum*

(2) *Selaginella*

(3) *Cycas*

(4) *Funaria*

15) Which of the following statment is incorrect in respect of members of kingdom monera ?

(1) They are simple in behaviour but complex in structure.

(2) Majority of bacteria are heterotrophs.

(3) Eubacteria play a great role in recycling of nutrients.

(4) Sequence of nucleotide in *16S-rRNA* of archaebacteria is differ from other bacteria.

16) Match column-I with column-II and select the correct option from the given codes :-

Column-I		**Column-II**	
(A)	Edible delicacies	(i)	*Penicillium*

(B)	*Drosophila* of plant kingdom	(ii)	*Ustilago*
(C)	Source of antibiotics	(iii)	*Neurospora*
(D)	Smut disease	(iv)	Morels

(1) A-ii, B-iii, C-i, D-iv

(2) A-iv, B-iii, C-i, D-ii

(3) A-ii, B-iv, C-i, D-iii

(4) A-iv, B-ii, C-iii, D-i

17) Choose the incorrect statement of following :-

(1) Ribosomes are the site of protein synthesis.

(2) Pili do not play a role in motility.

(3) Inclusion bodies are bounded by membrane.

(4) Cell wall prevent the bacterium from bursting and collapsing.

18) Some researchers were studying on a fungi that was unable to complete its life cycle on the dead decaying material, but it completes its life cycle on mustard that creates shiny white spots on leaves and stem. Which of the following is correct for the fungi they were studying?

(1) It is *Puccinia*, an obligate parasitic fungi.

(2) It is *Ustilago*, a facultative parasitic fungi.

(3) It is *Albugo*, an obligate parasitic fungi.

(4) It is *Agaricus* a facultative parasitic fungi.

19) Read the following statements :
(A) Prions cause Bovine Spongiform Encephalopathy
(B) Lichens do not grow in fresh air
(C) Viroids have ss RNA which is of high molecular weight
(D) In mycorrhiza, fungi increase the soil fertility by nitrogen fixation
Identify the option which is correct for above-mentioned statements :

(1) A and B are correct

(2) C and D are correct

(3) A, B and D are correct

(4) Only A is correct

20) Read the following statements carefully and identify the two statements which are not true and select the correct option from options given below :
(A) Branched and septate mycelium is present in members of ascomycetes, basidiomycetes and deuteromycetes.
(B) the asexual spores are endogenous and sexual spores are exogenous in most of members of basidiomycetes.
(C) Fruiting body in basidiomycetes is basidium.
(D) Asexual spores formed in imperfect fungi is conidium.

(1) A and D

(2) B and D

(3) A and B

(4) B and C

21) Match the features given in column-I with protistans given in column-II and select the correct option :

	Column-I		**Column-II**
(a)	Chief producer in Oceans	(i)	Slime moulds
(b)	Mesokaryon	(ii)	Diatoms
(c)	Mixotropic nutrition	(iii)	Dinoflagellates
(d)	Lack chlorophyll	(iv)	Euglenoids

(1) a-(ii), b-(iii), c-(iv), d-(i)

(2) a-(ii), b-(iii), c-(i), d-(iv)

(3) a-(i), b-(ii), c-(iii), d-(iv)

(4) a-(iv), b-(iii), c-(i), d-(ii)

22)

Karyogamy and meiosis take place in the basidium producing four basidiospores in

(1) *Alternaria*

(2) *Penicillium*

(3) *Trichoderma*

(4) *Ustilago*

23) Read the statements–
(a) Slime moulds form plasmodium during favourable conditions.
(b) Both Diatoms and Dinoflagellates are photosynthetic protist.
(c) Pigments of euglenoids are similar to higher plants.
(d) Spores in slime moulds have true cellulosic cell wall and formed during favourable conditions.
Choose the correct option:

(1) All statements are correct

(2) Only (a) and (c) are correct

(3) Only (a), (b) and (c) are correct

(4) Only (b) and (d) are correct

24) In which of the following the cell wall has stiff cellulose plate on the outer surface :-

(1) Slime moulds

(2) Chrysophytes

(3) Dinoflagellates

(4) Sporozoans

25) A : In diatoms Cell wall is made up of silica in which cellulose particles are embeded
B : Diatoms are photosynthetic
C : Cell wall of diatoms are indestructible due to presence of silica in cell wall.
D : Diatoms show active swimming movement at the surface of water.
E : Keiselgurh is involved in stone polishing, filteration of oil and syrup and in sound proofing.

(1) Statement A,B,C,D and E are correct
(2) Statement B,C,D and E are correct
(3) Statement A,B,C and E are correct
(4) Statement B,C, and E are correct

26) How many of the following produce sexual spores endogenously ?
Aspergillus, Trichoderma, Neurospora, Bracket fungi, *Colletotrichum, Claviceps, Alternaria, Penicillium*

(1) Three
(2) Four
(3) Six
(4) Seven

27) Which of the following is true for the given figure A and B ?

(A)  (B) 

(1) Sexual reproduction in A occurs by ascospores
(2) Sexual reproduction in B occurs by zygospores
(3) Sexual reproduction in A and B occur by ascospores
(4) Sexual reproduction in B occurs by conidia

28) Which of the following is the disease caused by the proteinaceous Infectious particles ?

(1) Small pox
(2) Potato leaf curl
(3) Tobacco mosaic disease
(4) Cr–Jacob disease

29) Select the correct option regarding viruses :

(1) AIDS, influenza, Tetanus
(2) Influenza, Mumps, smallpox
(3) Mumps, influenza, Aspergillosis

(4) Influenza, Small pox, Typhoid

30) Which of the following statement is incorrect with respect to lichens ?

(1) In Lichen algae component is known as Phycobiant and fungal component as mycobiant.
(2) Algae provide shelter to fungi.
(3) Very good pollution indicator.
(4) Fungi absorb mineral nutrients and water for its partner.

31) Read the following statements and find out the correct answer :
(A) *Porphyra* is edible algae.
(B) *Chlorella* and *Spirullina* are unicellular organisms.
(C) Algin is obtained from red algae.
(D) In pyrenoids starch is surrounded by protein.

(1) Statements A, C and D correct.
(2) Statement B and D correct
(3) Statement B and C correct
(4) Statements A and B correct

32) Choose the incorrect statement from following :-

(1) *Porphyra*, *Laminaria* and *Sargassum* are used as food
(2) Agar, a commercial products obtained from *Gelidium* and *Gracilaria*
(3) *Chlorella* is used as food supplement by space travellers
(4) Algin is obtained by red algae and carrageen is obtained by brown algae only

33) Read the following statement and choose the correct from following

(1) In *Ulothrix*, gametes can be non-flagellated and dissimilar in size
(2) In *Eudorina*, gametes are similar in size and flagellated
(3) In *Fucus* female gametes smaller and motile and male gametes larger and immotile.
(4) In *Spirogyra*, gametes are non-flagellated and similar in size

34) Sexual reproduction is oogamous and accompanied by "complex post fertilisation developments" in a group of algae in which :

(1) Spores and gametes are non-motile
(2) Dominance of chlorophyll a and c is observed
(3) Pyrenoids act as storage body commonly
(4) Vegetative cells are covered by gelatinous coating of Algin

35) "The sporophyte is not free living but attached to the photosynthetic gametophyte" above statement is related with :-

(1) Bryophyta
(2) Algae

(3) Gymnosperm

(4) Pteridophyte

36) Consider the following information :-
(a) Pyriform biflagellated male gametes
(b) Oogamous type of sexual reproduction
(c) Stored food - mannitol
Above informations are related to :-

(1) *Chara*

(2) *Fucus*

(3) *Cladophora*

(4) *Polysiphonia*

37) A group of algae, having following characters:-
(a) Presence of chlorophyll a and b
(b) Chloroplast has one or many pyrenoids
(c) Cell wall made of cellulose and pectose
(d) Reserve food material is starch
Which one of the following is also a very important character including above characters to call it as a member of chlorophyceae ?

(1) Plant body colonial only

(2) Plant body is made of haploid cells

(3) Its flagellar number 2–8, equal in length and apical in position

(4) It may occur in fresh water, brackish water or salt water

38) **Assertion :** The spread of living pteridophytes is limited and restricted to narrow geographical regions.
Reason : Prothallus of pteridophytes requires cool, damp, shady places to grow and need water for fertilization.

(1) Both assertion & reason are true & the reason is a correct explanation of the assertion.

(2) Both assertion & reason are true but reason is not a correct explanation of the assertion.

(3) Assertion is true but the reason is false.

(4) Both assertion & reason are false.

39) In fern, prothallus develops :

(1) By fertilization

(2) From zygote

(3) By secondary protonema

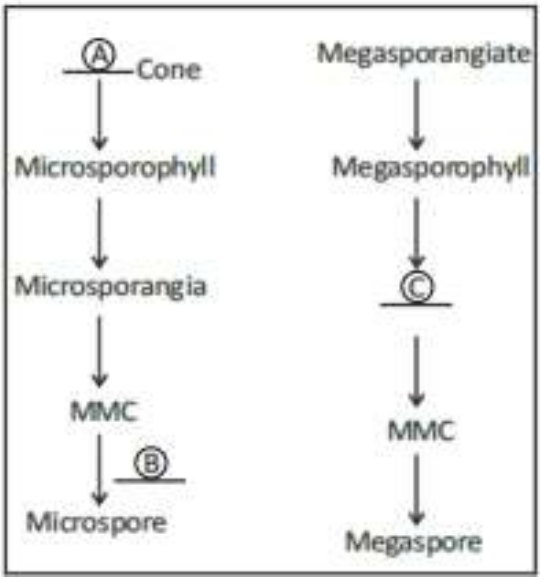
(4) From spore

40) Read the following terms :-
(i) Seed
(ii) Prothallus
(iii) Archegonium

(iv) Zygotic meiosis
(v) Sporophyll
How many terms are related to pteriodophytes

(1) 5
(2) 4
(3) 3
(4) 2

41) Complete following flow chart by using the options given below



(1) A – ♀, B – meiosis, C – megasporangium
(2) A – ♂, B – mitosis, C – nucellus
(3) A – ♀, B – meiosis, C – ovule
(4) A – ♂, B – meiosis, C – ovule

42) Choose the incorrect statement :-

(1) Female gametophyte of gymnosperm has two or more archegonia
(2) In Pinus male and female cones borne on the same tree but in Cycas male cones and mega sporophylls are borne on different trees
(3) Pollination in gymnosperms is carried by wind
(4) In gymnosperms male and female gametophytes have indepedent free- living existence

43) Match Column-I with Column-II :

	Column-I		**Column-II**
(a)	Giant red wood tree	(i)	*Cycas*
(b)	Mycorrhiza	(ii)	*Cedrus*
(c)	Coralloid roots	(iii)	*Sequoia*

(d)	Branched stem	(iv)	*Pinus*

Choose the correct answer from options given below.

(1) (a–i); (b–ii); (c–iii); (d–iv)

(2) (a–iii); (b–ii); (c–iv); (d–i)

(3) (a–iii); (b–iv); (c–i); (d–ii)

(4) (a–iv); (b–iii); (c–ii); (d–i)

44) Which of the following option is not related with *Cycas*?

(1) Unbranched stem

(2) Symbiosis with cyanobacteria

(3) More than one Archaegonia

(4) Monoecious sporophyte

45) **Assertion (A) :** Seed habit is present in the heterosporous members of pteridophyta. **Reason (R) :** The female gametophytes in heterosporous pteridophytes, are retained on the parent sporophytes for variable periods and the zygote develops in embryo within the female gametophytes.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

(3) **Assertion** is true but **Reason** is false.

(4) **Assertion** is false but **Reason** is true.

46) Which of the following animal is a homeiothermous ?

(1) Shark

(2) *Torpedo*

(3) *Petromyzon*

(4) *Rattus*

47) Which of the following animal has indirect development ?

(1) *Psittacula*

(2) *Petromyzon*

(3) *Betta*

(4) Sea horse

48) Which among these is the correct combination of viviparous animals ?

(1) *Macropus,* Platypus*, Camelus*

(2) *Platypus, Pteropus, Macaca*

(3) *Camelus, Macaca, Macropus*

(4) *Aptenodytes, Pteropus, Pristis*

49)

Gills are covered with operculum in

(1) *Scoliodon*
(2) *Myxine*
(3) *Catla*
(4) *Pristis*

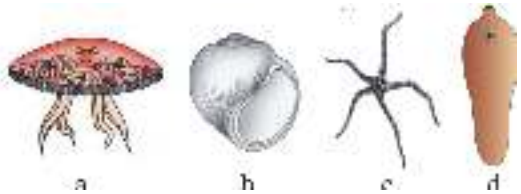
50) Which of the following animals is a true fish

(1) Star fish
(2) Sea horse
(3) Prawn
(4) Cuttle fish

51) Which group of animals belong to the same phylum ?

(1) *Pennatula*, *Gorgonia*, *Spongilla*
(2) *Pleurobrachia*, *Ctenoplana*, *Meandrina*
(3) *Ascaris*, *Ancylostoma*, *Wuchereria*
(4) *Nereis*, *Aedes*, *Apis*

52) Which of the following animals are radial symmetrical ?



(1) a & b
(2) b & d
(3) c & d
(4) a & c

53) **Assertion :** Platyhelminthes have incomplete digestive system.
Reason : Digestive system in Platyhelminthes has only one opening that serve as both mouth & Anus.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

54) Choose the correct statement :

(1) All mammals are viviparous
(2) All cyclostomates are ectoparasites on some fishes
(3) All reptiles have a three chambered heart

(4) All fishes have gills covered by an operculum

55) Read the following statements and choose the correct option
Statement 1: In Cartilaginous fish, gill slits are present (without operculum) and notochord is persistent throughout life.
Statement 2: In bony fish, air bladder is present which help in respiration and skin is covered with placoid scales.

(1) Both the statements are correct
(2) Both the statements are incorrect
(3) Only statement 1 is correct
(4) Only statement 2 is correct

56) Fibroblast, macrophages and mast cells are seen in :-

(1) Epithelial cells/tissues
(2) Loose connective tissue
(3) Skeletal muscle tissue
(4) Smooth muscle tissue

57) PCT of nephron is lined by :-

(1) Pseudostratified epithelium
(2) Stratified cuboidal epithelium
(3) Brush border cuboidal epithelium
(4) Non ciliated stratified columnar epithelium

58) Match column-I with column-II and select correct option :-

	Column-I		**Column-II**
(a)	*Cucumaria*	(i)	Hemichordata
(b)	*Aplysia*	(ii)	Aschelminthes
(c)	*Ancylostoma*	(iii)	Echinodermata
(d)	*Balanoglossus*	(iv)	Mollusca

(1) (a)-iii, (b)-ii, (c)-iv, (d)-i
(2) (a)-iii, (b)-iv, (c)-ii, (d)-i
(3) (a)-iv, (b)-iii, (c)-i, (d)-ii
(4) (a)-iv, (b)-iii, (c)-ii, (d)-i

59)

How many animals in the given list are respired through gills ?

Prawn, Labeo, Myxine, Bufo, Pteurobrachia, Adamsia, Hydra, Salpa, Trygon, Pavo, Apis
(1) Six

(2) Seven

(3) Nine

(4) Five

60) Choose the correct match from column-I, II and III:-

	Column-I		Column-II		Column-III
A		i	Platyhelminthes	a	Tissue level of organisation
B		ii	Aschelminthes	b	Organ system level of organisation
C		iii	Porifera	c	Organ level of organisation
D		iv	Coelenterata	d	Cellular level of organisation

(1) A-i-a, B-ii-b, C-iii-c, D-iv-d

(2) A-ii-b, B-iv-a, C-i-c, D-iii-d

(3) A-iii-c, B-i-d, C-ii-a, D-iv-a

(4) A-iv-d, B-iii-c, C-iv-b, D-i-b

61) **Statement-I :-** Echinoderms have an endoskeleton of calcareous ossicles and possess lateral appendages parapodia which help in swimming.
Statement-II :- Animals of Platyhelminthes are mostly endoparasites, they are triploblastic and pseudocoelomate animals.
Read the above statements carefully and select the correct option.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) Both **Statement I** and **Statement II** are correct.

(3) **Statement I** is correct but **Statement II** is incorrect.

(4) **Statement I** is incorrect but **Statement II** is correct.

62) In both male and female cockroach, which segment bears a pair of jointed filamentous structures called anal cerci ?

(1) 8th

(2) 9th

(3) 10th

(4) 7th

63) In cockroach head is formed by the fusion of :

(1) Six segments

(2) Three segments

(3) Four segments

(4) Nine segments

64) How many statements are not incorrect regarding class Cyclostomata.
(a) All living members of this class are endoparasites on some fishes.
(b) They have elongated body bearing 6-15 pairs of gill slits for respiration.
(c) Their body is devoid of scales & paired fins.
(d) They are fresh water species but migrate for spawning to marine water.
(e) Circulation is of closed type
(f) Cranium & vertebral column are bony.

(1) 5

(2) 4

(3) 3

(4) 2

65) Which of the following is incorrect option with respect to cockroach ?

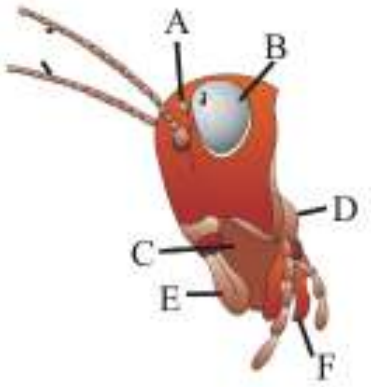
(1) The nymph grows by moulting about 13 times to reach the adult form

(2) Mushroom glands are accessory reproductive gland in male.

(3) The sperm are stored in the spermatheca and are glued together in the form of bundles called spermatophores.

(4) Ootheca is a dark reddish to blackish brown capsule, about 3/8" (8 mm) long

66) The figure given below shows the head region of cockroach. Identify A to F.



(1) A-Compound eye, B-Ocellus, C-Maxilla, D-Mandible, E-Labrum, F-Labium
(2) A-Ocellus, B-Compounds eye, C-Mandible, D-Maxilla, E-Labrum, F-Labium
(3) A-Ocellus, B-Compound eye, C-Mandible, D-Maxilla, E-Labium, F-Labrum
(4) A-Ocellus, B-Compound eye, C-Maxilla, D-Mandible, E-Labrum, F-Labium

67) In cockroach each ootheca consist of :

(1) 2000-4000 eggs
(2) 100-120 eggs
(3) 14-16 eggs
(4) 9-10 eggs

68) Choose the incorrect statement with respect to frog :-

(1) Vocal sacs are found only in male
(2) Copulatory pads are found in forelimbs of male frog
(3) Neck and nails are absent
(4) Hindlimbs have four digits and forelimbs have five digits

69) How do frogs primarily obtain oxygen when submerged in water ?

(1) Through the skin
(2) By swallowing air bubbles
(3) By breathing through lungs
(4) Through specialized gills

70) **Assertion :-** The mid brain of frog is characterised by a pair of optic lobes.
Reason :- The medulla oblongata passes out through the foramen magnum and continues into spinal cord.

(1) Both Assertion and reason are correct and reason is corret explanation of assertion.
(2) Both assertion and reason are true but reason is not a correct explanation of assertion.
(3) assertion is correct but reason is incorrect.
(4) Both assertion and reason are incorrect.

71) Which of the following is incorrect for scypha?

(1) It is primitive, acellular and asymmetrical animal

(2) It has a water transport system

(3) Digestion is intracellular

(4) Sexes are not separate

72) Match the following animals with their excretory organs :-

Animals		**Excretory organs**	
(A)	Earthworm	(i)	Rennete cell
(B)	Cockroach	(ii)	Nephridia
(C)	Taenia	(iii)	Malpighian tubules
(D)	Ascaris	(iv)	Flame cells

(1) A-ii B-i C-iv D-iii
(2) A-i B-ii C-iii D-iv
(3) A-ii B-iii C-iv D-i
(4) A-iii B-ii C-iv D-i

(1) 1

(2) 2

(3) 3

(4) 4

73) **Statement-I :** Members of cyclostomata are the chordate and are ectoparasite on some fishes.
Statement-II : Gnathostomata has two superclasses, Chondrichthyes & Tetrapoda.

(1) Statement-I & Statement-II incorrect.

(2) Statement-I correct & Statement-II incorrect.

(3) Statement-I & Statement-II both are correct.

(4) Statement-I incorrect but Statement-II correct.

74) Supplement respiration is found in which animal ?

(1) Chelone

(2) Calotes

(3) Columba

(4) *Balaenoptera*

75) Select the correct animal which show alternation of generation between Asexual and sexual form :

(1) sea fur (Obelia)

(2) sea anemone (Adamsia)

(3) sea cucumber (Cucumaria)

(4) Sea fan (Gorgonia)

76) Which one of the following categories of animals correctly described with no single exception in it ?

(1) All reptiles possess scales, have a three chambered heart and are cold-blood (poikilothermal)

(2) All bony fishes have four pairs of gills and an operculum on each side

(3) All sponges are marine and have collared cells

(4) All mammals are viviparous and possess diaphragm for breathing

77) Find out the incorrect statement :

(1) Cockroaches are serious pests and vectors of several diseases

(2) Cockroaches are nocturnal, herbivores animals

(3) Cockroaches are included in class insecta of phylum arthropoda

(4) Common species of cockroach is *Periplaneta americana*.

78) Match the following with respect to cockroach :-

Column-I		**Column-II**	
(A)	Ovaries	(i)	4th–6th
(B)	Testes	(ii)	2nd–6th
(C)	Mushroom gland	(iii)	14th–16th
(D)	Spermatheca	(iv)	6th–7th
(E)	Ootheca Number	(v)	6th segment
		(vi)	9–10

(1) A-(ii), B-(i), C-(iv), D-(v), E-(vi)

(2) A-(ii), B-(iv), C-(i), D-(iii), E-(v)

(3) A-(i), B-(iii), C-(ii), D-(v), E-(iv)

(4) A-(iii), B-(ii), C-(i), D-(iv), E-(v)

79) **Assertion :-** The body of the cockroach is segmented and the entire body is covered by a hard chitinous exoskeleton. **Reason :-** In each segment, exoskeleton has hardened plates called sclerites that are joined to each other by a thin and flexible arthrodial membrane.

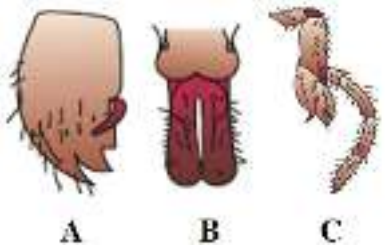
(1) **Assertion** is true but **Reason** is false.

(2) **Assertion** is false but **Reason** is true.

(3) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

(4) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

80) Identify the structure from figures A, B, C which is not correctly matched with their function.



(1) A – Mandible → Help in grinding.
(2) B – Labrum → Upper lip of cockroach.
(3) C – Maxilla → Form lateral wall of pre oral cavity.
(4) B – Hypopharynx → Tongue

81) If the spiracles of cockroach is blocked, what will be the effect of it :

(1) Air will not enter in trachea
(2) Blood will not enter in heart chambers
(3) Blood will not properly circulate in haemocoel
(4) Antenna of cockroach will not work properly

82) Which wings are transparent, membranous and are used in flight.

(1) Fore wings
(2) Meso thoracic wings
(3) Hind wings
(4) Both (1) and (2)

83) Male frog have :

(1) Copulatory
(2) Vocal Sac
(3) Both (1) and (2)
(4) Oviduct

84) How many vasa-efferentia arise from testes of male frog ?

(1) 25-30
(2) 10-12
(3) 10-pairs
(4) 3-4

85) Select the correct route for the passage of sperms in male frog in reverse order (outside to inside)

(1) Cloaca → Urinogenital duct → Seminal vesicle → Kidney → Vasa efferentia → Testes
(2) Cloaca → Ureter → Bidder's canal → Vasa efferentia → Testes
(3) Cloaca → Urinogenital duct → Bidder's canal → Kidney → Vasa efferentia → Testes

(4) Cloaca → Urinogenital duct → Vasa efferentia → Kidney → Bidder's canal → Testes

86)

The digestive system of frog consists of alimentary canal and digestive glands.

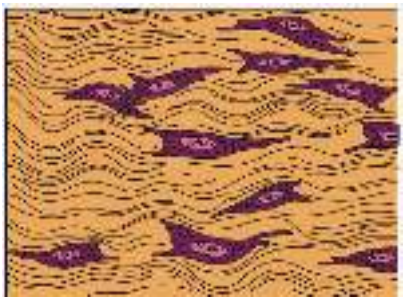
The alimentary canal is short because frog are carnivores. Pancreas, a digestive gland produces pancreatic juice containing digestive enzymes.
Statement-I :- Frog, are carnivores and hence length of intestine is reduced. **Statement-II :-** Final digestion take place in intestine digestive enzymes of bile juice and pancreatic juice digest carbohydrates and proteins.

(1) Both **Statement I** and **Statement II** are correct.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) Both **Statement I** and **Statement II** are incorrect.

(4) **Statement I** is incorrect but **Statement II** is correct.

87) 
The given figure represent :-

(1) Adipose tissue

(2) Areolar tissue

(3) Cartilage

(4) Ligament

88) Which cell junction prevent leaking of substances across the tissue ?

(1) Interdigitation

(2) Tight junction

(3) Adhering junction

(4) Gap junction

89) **Assertion :** Walls of blood vessels and air sac's of lungs have squamous epithelium.
Reason : Squamous epithelium involve in functions like forming a diffusion boundary.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

90) Read the following statements & choose correct for frog -
(i) Male reproductive organ consist of pair of yellow ovoid testes.

(ii) Cloaca is common opening for faecal, urine & sperm
(iii) Mature female can lay 2500 to 3000 ova at a time
(iv) Development shows larval stage.

(1) i, ii, iii, iv

(2) i, ii, iv

(3) i, & iv

(4) iii, iv

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	1	1	3	2	1	2	1	1	1	2	2	1	1	2	4	3	1	4	1	2
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	4	4	3	2	2	2	2	4	2	3	4	4	3	4	3	1	3	4	4	2
Q.	41	42	43	44	45															
A.	2	4	2	2	4															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	4	3	4	1	3	2	1	2	3	3	1	4	2	3	2	1	2	3	2	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	4	3	3	2	4	1	4	4	1	3	4	4	2	4	1	1	2	1	2	4
Q.	86	87	88	89	90															
A.	2	2	2	4	4															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	4	3	4	2	2	3	4	2	2	4	2	1	2	1	1	2	3	3	4	4
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	1	4	3	3	4	2	1	4	2	2	4	4	4	1	1	2	3	1	4	3
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	4	3	4	4	4	2	3	3	2	3	4	1	2	3	2	3	2	4	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	1	3	1	3	3	2	3	4	1	2	1	3	2	3	1	2	2	1	4	2
Q.	171	172	173	174	175	176	177	178	179	180										
A.	1	3	3	2	3	2	4	2	1	1										

SOLUTIONS

PHYSICS

1)

$u = -mx + u_0$

$\frac{du}{dt} = \frac{-mdx}{dt} + 0$

$\frac{du}{dt} = -mu = -m[-mx + u_0]$

$\boxed{a = +m^2x - mu_0}$ straight line

slope = +ve, intercept –ve

2)

$25h = 0 + \frac{1}{2} \times 10 \times t^2$

$25h = \frac{1}{2} \times 10 \times 5 \times 5$

$h = \frac{125}{25} = 5$ meter

in 1 second

$n \times 5 = 0 + \frac{1}{2} \times 10 \times (1)^2$

$n = \frac{5}{5}$

$\boxed{n = 1}$

3) $h = \frac{1}{2}gt^2$

$x = vt$

$t = \frac{x}{v}$

$h = \frac{1}{2}g\left(\frac{x}{v}\right)^2$

$\frac{h}{2h} = \left(\frac{x}{v}\right)^2 \times \left(\frac{v_2}{2x}\right)^2$

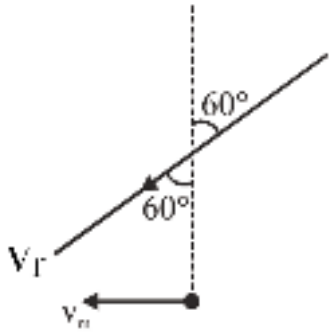
$\frac{1}{2} = \frac{v_2^2}{4v^2}$ $\boxed{v_2 = \sqrt{2}v}$

4) h = 20 – 12 = 8 m

$t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2 \times 8}{10}} = \sqrt{1.6}$

$x = u_x t \Rightarrow u_x = \frac{x}{t} = \frac{30}{\sqrt{1.6}} = \frac{30\sqrt{10}}{4}$

$\simeq 23.7 m/s$



5) $v_m = v_r \sin 60°$

$\Rightarrow v_r = \dfrac{v_m}{\sin 60°} \Rightarrow \dfrac{20 \times 2}{\sqrt{3}} = \dfrac{40}{\sqrt{3}} ms^{-1}$

6)

$\therefore x = 4t - 8 + at^2 + 4a - 4at$

$\therefore v = \dfrac{dx}{dt} = 4 + 2at - 4a$

$\therefore$ acceleration, $f = \dfrac{dv}{dt} = 2a$

7) $x = 9t^2 - t^3$

Finding position when maximum speed occurs (means acceleration is zero) :

$V = \dfrac{dx}{dt} = 18t - 3t^2$

$a = \dfrac{dV}{dt} = 18 - 6t$

For $a = 0$, $18 - 6t = 0 \Rightarrow t = 3$

At $t = 3s$, speed is maximum

then position, $x = 9 \times 3^2 - 3^3 = 54$ m

8)

$t = \alpha x^2 + \beta x$

$\dfrac{dt}{dt} = x\left[2\alpha \dfrac{dx}{dt}\right] + \beta \dfrac{dx}{dt}$

$1 = [2\alpha x + \beta]\dfrac{dx}{dt}$

$v = \left[\dfrac{1}{2\alpha x + \beta}\right]$

$a = \dfrac{dv}{dt}$

$a = -[2\alpha x + \beta]^{-2} \times 2\alpha \dfrac{dx}{dt}$

$a = \dfrac{-2\alpha}{[2\alpha x + \beta]^2} \times v$ $\boxed{a = -2\alpha v^3}$

9)

$\Rightarrow V = 4t - 3t^2$

$\Rightarrow \frac{ds}{dt} = 4t - 3t^2$

$\Rightarrow \int_0^s ds = \int_0^t (4t - 3t^2)\, dt$

$\Rightarrow S = 4\left[\frac{t^2}{2}\right]_0^t - 3\left[\frac{t^3}{3}\right]_0^t$

$\Rightarrow S = 2t^2 - t^3$

at t = 0, s = 0

at t = 2, $s = 2(2)^2 - (2)^3 = 0$

$V_{avg} = 0$

10)

$V_x = 4t$ | $V_y = 3$

$\int_0^x dx = 4\int_0^2 t\,dt$ | $\int_2^y dy = 3\int_0^2 dt$

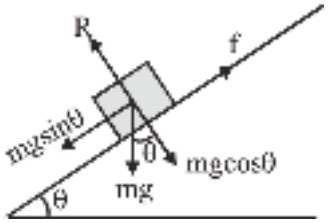
$x = 4\left[\frac{t^2}{2}\right]_0^2$ | $y - 2 = 3[t]_0^2$

$x = 4 \times 2 = 8$ | $y = 8$

So, at t = 2sec. | $\vec{v} = 8\hat{i} + 8\hat{j}$

11)

As is clear from figure.



R = mg cos θ

f = mR = μ mg cosθ

Component of weight (mg sin θ) acts down the incline. The body just begins to slide down when θ = $\tan^{-1}$ (μ), i.e., mg sin θ = μ mg cos θ. On increasing θ beyound the angle of repose (= $\tan^{-1}$μ), the body slides down with acceleration. Net contact force becomes

$F = \sqrt{(mg\cos\theta)^2 + (\mu mg\cos\theta)^2}$

$F = mg\cos\theta\sqrt{1+\mu^2}$

As θ increases, beyond angle of repose, contact force decreases and finally it becomes zero at θ = 90°.

12)

The forces exerted by both teams are equal because the rope is massless and cannot accelerate.

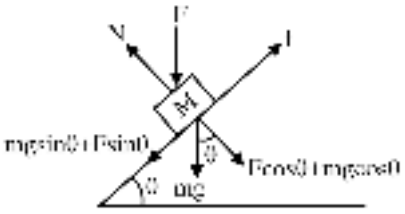
The tension is the rope is equal to the force exerted by the teams which is consistent with the

conclusion from statement-1.

13)

Direction of $\vec{F}_1$ & $\vec{F}_2$ is different.

14)

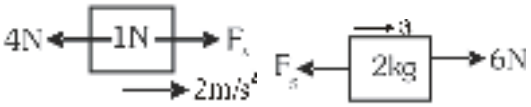


$F = \mu_k N$

$N = (F\cos\theta + mg\cos\theta)$

$F = \mu_k (F + mg)\cos\theta$

15)



$F_S - 4 = 2$ $\qquad$ $6 - F_S = 2 \times a$

$F_S = 6$ $\qquad$ $6 - 6 = 2 \times a$

$a = 0$

16) $\frac{dM}{dt}$ = 0.1 kg/s, v_{gas} = 50 m/s,

mass of the rocket = 2 kg, Mv = constant

$$F = v\frac{dm}{dt}$$

or acceleration = $\frac{1}{2} \times 50 \times 0.1 = 2.5\ m/s^2$.

17) $a_1 = \frac{F}{m_1},\ a_2 = \frac{F}{m_2}$

$$a = \frac{F}{m_1 + m_2} = \frac{F}{\frac{F}{a_1} + \frac{F}{a_2}} = \frac{a_1 a_2}{a_1 + a_2}$$

$$a = \frac{3 \times 6}{3 + 6} = 2 m/s^2$$

18)

Static friction adjusts up to $f_s^{max} = \mu_s N$.

Normal reaction adjusts based on external forces (e.g., weight and inclination).

Tension in a string adjusts according to applied forces until the string becomes taut or reaches its breaking strength.

19)

Impulse = $m(\vec{v}_f - \vec{v}_i)$

$-3m\hat{i} = m(\vec{v}_f - \vec{v}_i)$

$\vec{v}_f = -3\hat{i} + \vec{v}_i$

$= -3\hat{i} + (2\hat{i} + 3\hat{j})$

$= \left(-\hat{i} + 3\hat{j}\right)\ m/sec.$

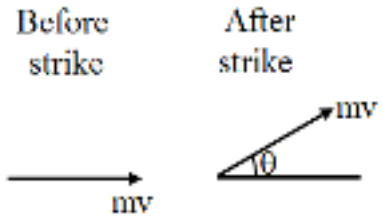
20)

V from 0 to 2 is $\frac{dx}{dt} = 1 m/s$

V from 2 to 4 is $\frac{dx}{dt} = -1 m/s$

$J = 0.2(1 - (-1))$

$J = 0.4 NS$



21)

$|\vec{P}_i| = |\vec{P}_f| = mv$

$|\Delta\vec{P}| = 2mv \sin\frac{\theta}{2}$

22)

Total mass: 5 + 10 = kg

Acceleration: $a = \frac{F}{15}$

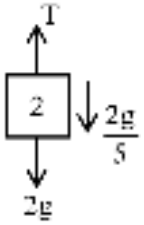
Maximum static friction: $5 \times \frac{F}{15} \leqslant 15$

$F \leqslant 45N$

23) $a = \frac{F}{M}$

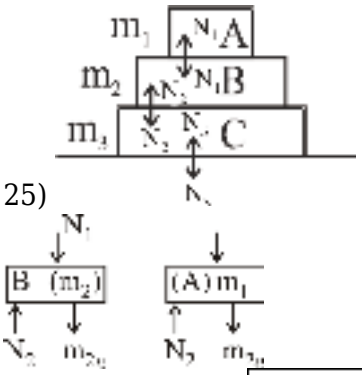
24)

$a = \frac{2g}{2+3} = \frac{2g}{5}\ m/s^2$



$$T = M\left(g - \frac{2g}{5}\right) = \frac{6g}{5}N$$

25)



$N_2 = N_1 + m_2g$ $\boxed{N_1 = m_1g}$

$N_2 = m_1g + m_2g$

$\boxed{N_2 = (m_1 + m_2)g}$

26)

Due to inertia of rest of block jerk is not transferred to upper cord.

27) Relative acceleration of blocks with respect to pulley

$$a_{rel} = \frac{m_2 - m_1}{m_2 + m_1}(g + a_p)$$

$$= \frac{10 - 5}{10 + 5} \times (10 + 5)$$

$$= \frac{5}{15} \times 15$$

$$= 5\ m/s^2$$

28)

$mg\sin\theta - T = ma$

$T = ma$

$mg\sin\theta = 2ma$

$$a = \frac{g\sin\theta}{2}$$

$T = ma$

$$= \frac{mg\sin\theta}{2}$$

29)

For stable equilibrium

$$\frac{\partial F}{\partial x} < 0$$

at point x_2, $\frac{\partial F}{\partial x} < 0$
so at $x = x_2$ body is in stable equilibrium

30) $a = \frac{F}{m}$
$v = u + at = at$
$S = \frac{1}{2}at^2 \; ; \; t^2 = \frac{2s}{a}$
$KE = \frac{1}{2}mv^2 = \frac{1}{2}m \times a^2 \times t^2$
$\boxed{KE \propto t^2}$
$P = Fv = F(at) = Fat$
$KE = \frac{1}{2}ma^2t^2 = \frac{1}{2}ma^2 \times \frac{2s}{a}$
$KE = maS$
$KE = FS$
$\boxed{KE \propto S}$

31) $U_1 = \frac{1}{2}kx^2$
$U_2 = \frac{1}{2}k(x+y)^2$
$W = U_2 - U_1$
$= \frac{1}{2}k(x+y)^2 - \frac{1}{2}kx^2$
$= \frac{1}{2}k(2xy + y^2)$
$\boxed{W = \frac{1}{2}ky(2x+y)}$

32)

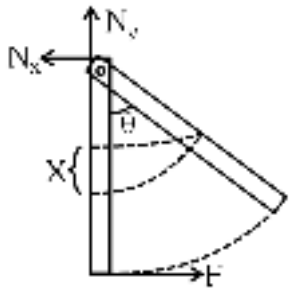
By, $v = u + at$
$\frac{v}{2} = 0 + at$
$a = \frac{v}{2t}$ .....(1)

$$P_{avg} = \frac{w}{t} = \frac{\frac{1}{2}m\left(\frac{v}{2}\right)^2}{t} = \frac{1}{4}\left(\frac{mv}{2t}\right)\cdot v$$
$$= \frac{Fv}{4}$$

33)

Loss in PE = gain in KE
$Mgh = \frac{1}{2}mv^2 + \frac{1}{2}Mv^2$

$$v = \sqrt{\frac{2Mgh}{M+m}}$$

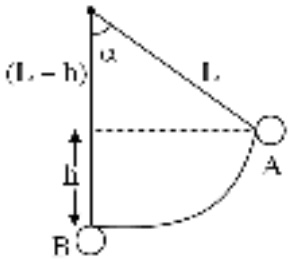


34)

$$W_g + W_{N_X} + W_{N_Y} + W_F = \Delta KE$$

$$-mg\left(\frac{\ell}{2} - \frac{\ell}{2}\cos 60^o\right) + 0 + 0 + W_F = 0$$

$$W_F = mg\frac{\ell}{4} = 12.5\ J$$



35)

h = L(1 – cos α)

PE at A = KE at B

$MgL\ (1 - \cos\alpha) = K_B$

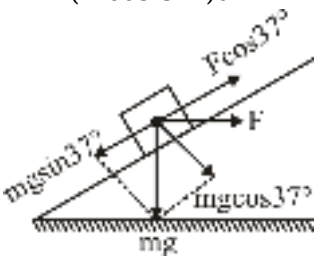
36)

As velocity is constant

Fcos 37° = mgsin 37°

So WD by F

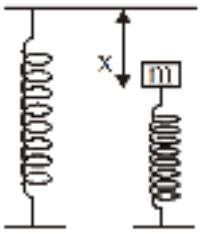
W = (F cos 37°)d



= (mg sin 37°)d

$= 50 \times \frac{3}{5} \times 2$

= 60 J

37) 

$mg(h+x) = \frac{1}{2}kx^2 \Rightarrow k = \frac{2mg(h+x)}{x^2}$

38) $v = 2x \Rightarrow \frac{dx}{dt} = 2x$

$\Rightarrow \int_2^x \frac{dx}{x} = \int_0^t 2dt$

$\Rightarrow \ln x - \ln 2 = 2t$

$\Rightarrow \ln\frac{x}{2} = 2t \Rightarrow x = 2e^{2t}$

39) $\vec{F} = -\frac{\partial U}{\partial x}\hat{i} - \frac{\partial U}{\partial y}\hat{j} = 7\hat{i} - 24\hat{j}$

$\vec{a} = \frac{7}{5}\hat{i} - \frac{24}{5}\hat{j}$

$\vec{v} = \vec{a}.t = \frac{14}{5}\hat{i} - \frac{48}{5}\hat{j}$

$|\vec{v}| = \frac{1}{5}\sqrt{14^2 + 48^2} = 10\,m/sec$

40) $W = \int_{(0.0)}^{(1.1)} \vec{F}.d\vec{s}$

Here $\vec{ds} = dx\hat{i} + dy\hat{j} + dz\hat{k}$

$W = \int_{(0,0)}^{(1,1)} (x^2dy + ydx) = \int_{(0,0)}^{(1,1)} (y^2dy + xdx)$

as (x = y)

$W = \left[\frac{y^3}{3} + \frac{x^2}{2}\right]_{(0,0)}^{(1,1)} = \frac{5}{6}J$

41)

By COME law

$mE_i = mE_f$

$-\frac{m}{3}g\frac{L}{6} + 0 = -mg\frac{L}{2} + \frac{1}{2}mv^2$

$v^2 = \frac{8Lg}{9} \Rightarrow v = \frac{2}{3}\sqrt{2gL}$

42) Least count (LC) = $\left(\frac{b-a}{b}\right)$MSD;

here a = 9, b = 10;
1MSD = 1mm
LC = $\left(\frac{10-9}{10}\right)$ 1mm = 0.1mm
There is zero error in measurement.
As the zero of the vernier lies to the left of the main scale, it has negative zero error.
For negative zero error, the coinciding division is to be read from right and it has 2nd division coinciding from right. Thus zero error
= ZE = –2 VSD = –2 × 0.1mm
= –0.2mm
Reading at measurement: 11MSD + 6VSD (coinciding)
Corrected reading = MSR + VSR × LC – zero error = 11mm + [6 × 0.1mm] – (–0.2mm) = 11.8mm
Hence the value of x = 118

43) Pl = 0.5 mm
N = 100 divisions
zero error = 2 divisions (+ve error)
Reading = Measured value + zero correction
= (8 × 0.5) mm + (83 – 2) × $\frac{0.5}{100}$.
= 4 mm + 81 × $\frac{0.5}{100}$ mm
= 4.405 mm

44) Least count of screw gauge
= $\frac{\text{Pitch}}{\text{no. of division on circular scale}}$
= $\frac{0.5}{50}$mm = 1×10^{-5}m
= 10 µm
Zero error in positive
Ans. (2)

45) □ = 3.50 cm
So *LC* of instrument is 0.01 cm
(1) *LC* = $\frac{1\text{mm}}{100}$ = 0.01 *mm*=0.001 *cm*
So it can't be the answer
(2) *LC* = $\frac{1\text{mm}}{50}$ = 0.02 mm = 0.002 cm
(3) Meter scale has 1 *cm* as *LC*
(4) *LC* = 1 *MSD* – 1 *VSD* $\left[\begin{array}{l} 10\text{VSD} = 9\text{MSD} \\ 1\text{VSD} = 0.9\text{MSD} \end{array}\right]$
= 1 *MSD* – 0.9 *MSD*
= 0.1 *MSD*
and 1 *MSD* = $\frac{1\text{cm}}{10}$ = 0.1 *cm*

 $LC = 0.01\ cm$ So this is correct option.

CHEMISTRY

46) **A. Question**
The equivalent weight of phosphoric acid (H_3PO_4) in the reaction,
$NaOH + H_3PO_4 \rightarrow NaH_2PO_4 + H_2O$, is:
B. Given Data
• Reaction: $NaOH + H_3PO_4 \rightarrow NaH_2PO_4 + H_2O$
• Phosphoric acid (H_3PO_4) is a triprotic acid, but in this reaction, only one proton is neutralized (as indicated by the formation of NaH_2PO_4)**.**
E. Final Answer
The equivalent weight of H_3PO_4 in this reaction is 98.
(Option 4)

47) $CO_{2(g)} + C \rightarrow 2CO_{(g)}$
1L – -
1 – x 2x
Total ⇒ 1 – x + 2x = 1.5L ⇒ (x = 0.5L)
$V_{CO} = 2x = 1L$ & $n_{CO} = \frac{1}{22.4}$ mole at STP

48)

No. of O - atom = $\frac{8}{6} = \frac{4}{3}$
for $Ca_3(PO_4)_2$
$\frac{no.of P-atom}{no.of O-atom} = \frac{2}{8}$
for H_3PO_4
$\frac{no.of P-atom}{no.of O-atom} = \frac{1}{3} = \frac{2}{6}$

49) Mole of $O_2 = \frac{67.2}{22.4} = 3$ mole
$3O_2 \rightarrow 2O_3$
 Mole of Ozone formed = $\frac{2}{3} \times \frac{15}{100} \times 3$
= 0.3 mole
 Mass of ozone formed = 0.3 × 48g = 14.4 g

50) $M_2O_3 + 3H_2 \rightarrow 2M + 3H_2O$
gram eq. of M_2O_3 = gram eq. of H_2
$\frac{0.1595}{E_M + 8} = \frac{6 \times 10^{-3}}{1}$
$E_{metal} = 18.583$
At wt of metal = 3 × 18.583

= 55.75

51) Suppose, the volume of propane = VL

$C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(l)$

V 5V 3V

$C_4H_{10}(g) + \frac{13}{2} O_2(g) \rightarrow 4CO_2(g) + 5H_2O(l)$

(3-V) $\frac{13}{2}$ (3 -V) 4(3-V)

Total volume of CO_2 produced = 10 L;

3V +4 (3 – V) = 11; V = 1

Volume of butane = (3 –1) = 2L

Thus, the ratio of volume of butane to propane = 2 : 1

52)

$$\text{V.F.} = \frac{2 \times \text{V.D}}{\text{E} + 35.5} = 2.66$$

Atomic wt. = 2.66 × 9 = 23.9

53) $\frac{50.35}{100} = \frac{142}{142 + 18x}$

x = 8

54)

$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$

$\left(n_{Fe_2O_3} = \frac{480}{160} = 3\text{mol}\right)$

$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$

(3mol) (excess)

O 6 mol 9 mol

Theoretical yield = 6 mol = 336 gm of Fe

Actual yield = 4.5 mol = 252 gm

$$\% \text{ yield} = \frac{\text{Actual yield}}{\text{Theoretical yield}} \times 100 = \frac{252}{336} \times 100 = 75\%$$

55) No. of Fe atoms = $\frac{0.34}{100} \times \frac{3.3}{56} \times 6.022 \times 10^{23}$

= 1.206×10^{20}

56) **Question Explanation:** We need $\mathbf{K_p}$ for the equilibrium reaction at 900 K. At equilibrium, the gaseous mixture has 45% H_2 by volume, so mole fractions are directly used. Solids and liquids are not included in K_p.

Concept: Volume % = mole fraction for gases, Equilibrium constant (K_p)

Solution:

K_p expression depends only on gaseous species.
For the reaction: $SnO_2(s) + 2H_2(g) \rightleftharpoons 2H_2O(g) + Sn(l)$

$$K_p = \frac{(P_{H_2O})^2}{(P_{H_2})^2}$$

Mole fraction of H_2 = 0.45
Mole fraction of H_2O = 0.55

So, $P_{H_2O} = 0.45P$ $P_{H_2O} = 0.55P$

$$K_p = \frac{(0.55P)^2}{(0.45P)^2}$$

$= (0.55 / 0.45)^2$
$= (11 / 9)^2$
= 1.49
Final Answer: Option (1)

57)

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Q_P for given reaction = $\frac{P_{CO_2}}{P_{CO}} = \frac{0.8}{1.6} = \frac{1}{2} = 0.5$

$\therefore Q_P > K_P$ hence the reaction will proceed in the backward direction

$Fe(s) + CO_2(g) \rightleftharpoons FeO(s) + CO(g)$;

$$K_p' = \frac{1}{K_P} = \frac{1}{0.25} = 4$$

t = 0	0.8 atm	1.6 atm
$t = t_{eq}$	(0.8 – x)	(1.6 + x)

$$K'_P = \frac{P_{CO}}{P_{CO_2}} \Rightarrow 4 = \frac{(1.6 + x)}{(0.8 - x)}$$

$\Rightarrow 3.2 - 4x = 1.6 + x \Rightarrow 5x = 3.2 - 1.6 = 1.6$
$x = = 0.32$
Hence equilibrium partial pressure of CO(g)
= 1.6 + 0.32 = 1.92 atm
equilibrium partial pressure of $CO_2(g)$
= 0.8 – 0.32 = 0.48 atm

58) $A + D \rightleftharpoons AD$, K_1 ..... (1)
$AD + D \rightleftharpoons AD_2$, K_2 ..... (2)
$AD_2 + D \rightleftharpoons AD_3$, K_3 ..... (3)
$A + D_3 \rightleftharpoons AD_3$, K .....(4)
(4) = (1) + (2) + (3)
$k = k_1 \times k_2 \times k_3$
$\log k = \log k_1 + \log k_2 + \log k_3$

59) $N_2O_3 \rightleftharpoons NO + NO_2$

1	–	–

$$\frac{1-\alpha}{1+\alpha}P \quad \frac{\alpha}{1+\alpha}P \quad \frac{\alpha}{1+\alpha}P$$

$$\Rightarrow K_p = \frac{\alpha^2 P}{1-\alpha^2} = \frac{0.2}{1-0.2} \times 10 = 2.5 \text{ atm}$$

60)

Question Explanation : Extra A(g) is added to equilibrium A(g) + B(g) ⇌ C(g) + D(g). We must pick the graph showing rate changes until a new equilibrium is established.
Concept : This question is based on Le-Chatelier's principle (effect of concentration on equilibrium rates)
Solution : Adding A(g) increases forward rate (r_f). Backward rate (r_b) remains unchanged initially. With time, r_b increases until $r_f = r_b$ again at new equilibrium. This matches graph (2).

Final Answer : option (2)

61)

Step-by-Step Calculation:
1. Assume initial moles of N_2O_3 = 1 mole.
2. Degree of dissociation $\alpha = \sqrt{0.5} \approx 0.707$.
3. At equilibrium: - Moles of $N_2O_3 = 1-\alpha$ - Moles of NO = α - Moles of $NO_2 = \alpha$
4. Total moles = $1-\alpha+\alpha+\alpha = 1+\alpha$.
5. Partial pressures (total pressure = P):

$$p_{N_2O_3} = \frac{1-\alpha}{1+\alpha}P, \quad p_{NO} = \frac{\alpha}{1+\alpha}P, \quad p_{NO_2} = \frac{\alpha}{1+\alpha}P$$

6. Calculate Kp:

$$K_p = \frac{p_{NO} \times p_{NO_2}}{p_{N_2O_3}} = \frac{\left(\frac{\alpha}{1+\alpha}P\right)^2}{\frac{1-\alpha}{1+\alpha}P} = \frac{\alpha^2 P}{1-\alpha^2}$$

7. Since $\alpha^2 = 0.5$ and $1-\alpha^2 = 0.5$, $K_p = \frac{0.5P}{0.5} = P$ Hence, Kp equals the total pressure.

62)

<u>Explanation</u>

A. This is a classic acid-base titration problem involving a weak acid (acetic acid) and a strong base (NaOH).
B. The goal is to find the volume of NaOH needed to reach a specific pH, which indicates the formation of a buffer solution.
C. The pH of 4.74 is significant because it's very close to the pK_a of acetic acid, meaning we're aiming for a buffer where the concentrations of the acid and its conjugate base are nearly equal.

<u>Given Data</u>

A. [NaOH] = 0.2 M

B. [CH_3COOH] = 0.2 M, 50 mL
C. Ka (CH_3COOH) = 1.8×10^{-5}
D. Desired pH = 4.74

Concept

A. **Henderson-Hasselbalch Equation:** pH = pK_a + log([A^-]/[HA])
    A. Where:
        A. pH is the desired pH.
        B. pK_a = -log(Ka).
        C. [A^-] is the concentration of the conjugate base (acetate ion).
        D. [HA] is the concentration of the weak acid (acetic acid).
B. When the pH=pK_a, then [A^-]/[HA]=1. Therefore, [A^-]=[HA].

Calculation

A. **Calculate pK_a:**
    A. pK_a = -log(1.8×10^{-5}) ≈ 4.74
B. **Apply Henderson-Hasselbalch:**
    A. Since the desired pH (4.74) equals the pK_a (4.74), the concentration of acetate ion (A^-) must equal the concentration of acetic acid (HA) in the buffer solution.
    B. This means that half of the acetic acid has been neutralized by the NaOH.
C. **Calculate NaOH volume:**
    A. Since the concentrations of NaOH and acetic acid are the same, the volume of NaOH needed is half the initial volume of acetic acid.
    B. Volume of NaOH = 50 mL / 2 = 25 mL

Answer option 2, (25 ml).

63) $\underset{s}{BaF_2} \rightleftharpoons \underset{s}{Ba^{+2}} + \underset{2s}{2F^-}$
2s = [F^-]
$s = \frac{[F^-]}{2}$

64) Enthalpy of ionization $\propto \frac{1}{\text{strength of acid}}$
strength of acid $\propto \frac{1}{\text{pka value}}$

65) $K_{sp} = (S)\,(0.2)^2$
$\Rightarrow S = 0.45 \times 10^{-9}$

66) $\left(Pb^{+2}\right)\left(SO_4^{-2}\right) = K_{sp}$
$\left[Pb^{+2}\right]\left[45 \times 10^{-4}\right] = 1.8 \times 10^{-8}$
$\left[Pb^{+2}\right] = 4 \times 10^{-6} M$

67)

	NH_4OH	+ HCl	→ NH_4Cl	+ H_2O
t = 0	5	4	0	0
t = eq	1	0	4	4

Mixture of (NH_4OH + NH_4Cl) is basic buffer solution.

68)

	CH_3NH_2	+ HCl →	CH_3NH_3Cl
initial milli-moles	0.12	0.08	
after reaction	0.04	—	0.08

$$pOH = pK_b + \log \frac{[CH_3NH_3^+]}{[CH_3NH_2]}$$

$$= 3.3 + \log \frac{0.08}{0.04} = 3.6$$

∴ pH = 10.4

69) [Salt] = [Acid]

70)

$$pH = pK_a + \log \frac{[salt]}{[acid]}$$

$$4.75 = 4.87 + \log \frac{[salt]}{[0.02]}$$

$$\log \frac{salt}{0.02} = -0.122$$

[salt] = 0.755 × 0.02

= 1.52 $\times 10^{-2}$M

71) **Explanation :** With the help of Q_{SP} and K_{sp}, we have to find out whether the solutin will precipitate or not.

Given Data :

$K_{sp} = 8.5 \times 10^{-7}$

$[Ag^+] = 0.010$ M

$[I^-] = 0.015$ M

Concept : $Q_{sp} > K_{sp}$ (solution will precipitate)

Calculation :

$AgI \rightleftharpoons Ag^+ + I^-$

$Q_{sp} = [Ag^+]\,[I^-] = 0.10 \times 0.015$

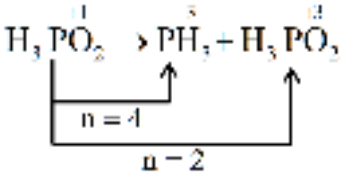
$Q_{sp} = 1.5 \times 10^{-4}$ $Q_{sp} > K_{sp}$

72)

pH = WAWB

$$= 7 + \frac{1}{2}pK_a - \frac{1}{2}pK_b$$

= 7 ∴ pH = 7

73) 

$$n = \frac{n_1 n_2}{n_1 + n_2} = \frac{4 \times 2}{4 + 2} = \frac{8}{6} = \frac{4}{3}$$

$$\text{Eq. wt.} = \frac{M}{4/3} = \frac{3M}{4}$$

74) $4OH^- + 2Cr(OH)_3 + IO_3^- \rightarrow 5H_2O + 2CrO_4^{2-} + I^-$

75) Lose of electron for 1 mol of $N_2H_4 = \frac{25}{2.5} = 10$
Lose of electron for per nitrogen atom = 5

$$N_2^{-2}H_4 \rightarrow N^{x+} + 5e^-$$

x = –2 + 5 = +3

76)

gram eq ($KMnO_4$) = 1 × 5 = 5
gram eq (Fe SO_4) = n_1 × 1
n_1× 1 = 5
n_1 = 5
gram eq (Fe C_2O_4) = n_2 × 3
n_2 × 3 = 5
n_2 = 5/3
gram eq ($H_2C_2O_4$) = N_3 × 2
N_3 × 2 = 5

$$N_3 = \frac{5}{2} = 2.5$$

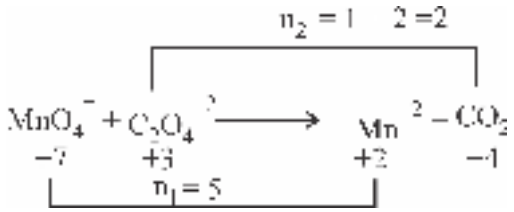
77)

In acidic medium valence factor of $KMnO_4$ = 5
($MnO_4^- \rightarrow Mn^{2+}$) and $X^{n+} \rightarrow XO_3^-$ thus valence factor of X^{n+} = 5 – n
Now, from the law of equivalence,
eq.($KMnO_4$) = eq. (X^{n+})
⇒ moles × valence factor = moles × valence factor
⇒ $6 \times 10^{-3} \times 5 = 9 \times 10^{-3} \times (5-n)$

$$\Rightarrow \frac{30}{9} = (5 - n)$$

$$\Rightarrow n = 5 - \frac{30}{9} = \frac{15}{9} = \frac{5}{3}$$

78)



(nes)MnO_4^- = (nes) $C_2O_4^{-2}$

x×5 = 2×y

$\frac{x}{y} = 0.4$

79) Weak acid release less energy on neutralization with S.B. and thus weak acid has strong conjugate base which has max^m degree of hydrolysis.

80)

$CaO(s) + CO_2(g) \rightarrow CaCO_3(s)$

$\Delta H = (\Delta H)_P - (\Delta H)_R$

$\Delta H = -178$ kJ/mol

$\Delta H = \Delta E + \Delta ngRT$

81)

From Hess law

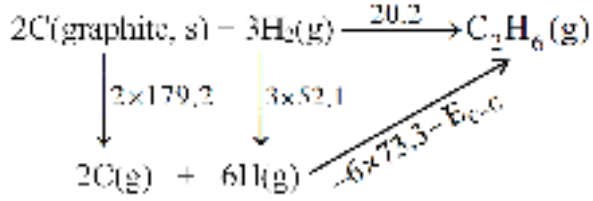
ΔH = Eq.(1) + Eq.(2) + Eq.(3) – Eq.(4)

82)

Heat of formation of H_2O equation is

$H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow H_2O_{(\square)}$; $\Delta H = -X_2$ kJ mol^{-1}

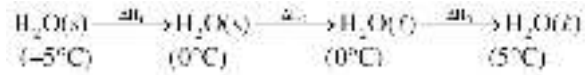
83)



$-20.2 = 2 \times 179.2 + 3 \times 52.1 - 6 \times 73.3 - E_{C-C}$

E_{C-C} = 95.1 kCal/mol

84)



$\Delta H = \Delta H_1 + \Delta H_2 + \Delta H_3$

Here $\Delta H_1, \Delta H_2$ and ΔH_3 all are positive

85) NCERT-XI, Pg # 183, Q.No. 6.15

86)

$\Delta H = \Delta E + (P_2V_2 - P_1V_1)$
$\Delta H = \Delta E + V(P_2 - P_1)$
$-543 = \Delta E + 1(40 - 70)$
$-543 = \Delta E - 30$ atm L
$-543 = \Delta E - \frac{30 \times 100}{1000}$
$\Delta E = -543 + 3$
$= -540$ kJ mol^{-1}

87) $\Delta H_{surr} = -\Delta H_{sys}$
$\Delta S_{surr} = \frac{+\Delta H_{surr}}{T} = \frac{-\Delta H_{sys}}{T}$
$= \frac{-2000}{400} = -5 \frac{KJ}{mol\ K}$

88) $\Delta H = \Delta E + \Delta ng\ RT$

89)

$H_2NCOONH_4(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$
$2x \quad x$
$K_p = (2x)^2x = 4x^3$
$\Delta G° = -RT \ln K_p$
$\Delta G° = -RT \ln 4x^3$
$= -RT \ln 4 - 3RT \ln x$

90)

NCERT pg # 166
$\Delta s = nR \ln \frac{V_2}{V_1}$

BIOLOGY

91) NCERT-XI Pg. # 23, 24

92) NCERT-XI Pg. # 7

93) NCERT-XI Pg. # 6, 7

94)

NCERT-XI Pg. # 6

95)

NCERT-XI Pg. # 24

96)

NCERT-XI, Pg. # 4

97)

NCERT-XI, Pg. # 7

98)

NCERT-XI, Pg. # 5, 6

99)

NCERT-XI, Pg. # 11

100)

NCERT-XI, Pg. # 11

101) NCERT (XI) Pg. # 90

102) NCERT-XI, Pg. # 12, 13

103)

NCERT-XI, Pg. # 13, 14, 91

104) NCERT-XI, Pg. # 29, 30

105) NCERT XI (E) Pg. # 12

106)

NCERT-XI, Pg. # 17, 18

107)

NCERT-XI, Pg. # 90, 91

108)

NCERT-XI, Pg. # 17

109) NCERT XI Pg. # 21

110) NCERT XI Pg. # 17, 18

111)

NCERT-XI, Pg. # 14, 15

112)

NCERT-XI, Pg. # 18

113) NCERT-XI, Pg. # 14, 15
Only (a), (b) and (c) are correct

114) NCERT-XI Pg. # 15

115)

NCERT XI, Pg. # 14

116) NCERT XI Pg. # 17 and 18
(*Aspergillus, Neurospora, Claviceps, Penicillium produce endogenous spores.*

117) NCERT XI Pg. # 17

118)

NCERT-XI, Pg # 21

119)

NCERT-XI, Pg. # 14

120) NCERT, Pg. # 21

121)

NCERT-XI, Pg. # 26

122) NCERT Pg. # 26

123) NCERT Pg. # 26

124)

NCERT XI, Pg. # 27

125) NCERT-XI, Pg. # 28

126) NCERT-XI, Pg. # 27

127)

NCERT XI, Pg. # 26

128) NCERT XI, Pg. # 36

129) NCERT Pg. # 36, 37

130)

NCERT Pg. # 36

131)

NCERT XI, Pg. # 33

132)

NCERT XI, Pg. # 33

133) NCERT-XI Pg. # 32, 33

134)

NCERT-XI, Pg. # 33

135)

NCERT Pg. # 32

136) NCERT Page-60

137) NCERT Page-56

138)

NCERT XI, Pg. # 51

139) NEERT Page No. # 57

140) NEERT Page No. # 57

141) NEERT Page No. # 52

142) NCERT Pg. # 50,53

143)

NCERT Pg. # 47

144)

NCERT-XI, Pg. # 47

145) NCERT Pg. # 56, 57

146) NCERT (XIth) Pg. # 103

147) NCERT Pg. # 101, para 2

148) NCERT (XIth) Pg. # 52,54

149) NCERT XI (E) Pg. # 40 to 50

150) NCERT (XI) Pg. # 40,41,42,43

151)

NCERT Pg. # 43, 45

152) NCERT, Page # 112(E), 113(H)

153) NEERT Page No. # 112

154) NCERT Page No. # 56

155) NCERT-XI, Pg. # 114,115

156)

A-Ocellus, B-Compounds eye, C-Mandible, D-Maxilla, E-Labrum, F-Labium

157)

NCERT XI Pg # 115

158) NCERT Pg # 80-81

159)

NCERT-XIth, Pg. # 117-118

160)

NCERT Pg.#119

161) Pg. No. 49 NCERT 2022 - 2023 Edition

162)

NCERT Pg. # 43,44

163) NCERT Pg. # 61

164) NCERT XI Pg. # 59

165)

NCERT Pg. No. # 50

166) NCERT-XI, Pg. # 48

167) NCERT Pg.#111 (E) 112 (H)

168)

NCERT-XI, Pg. # 114,115

169)

Module

170) NCERT-XI Pg. # 112

171) NCERT Pg.#113 (E), 114 (H)

172) NCERT-XI, Pg # 112, Ist PARA

173)

NCERT Pg # 116

174) New XI, NCERT, Pg. # 83

175) NCERT Pg. # 83

176)

NCERT Pg. # 81

177) NCERT (XI) Pg. # 103

178) NCERT (XIth) Pg. # 102, II para

179) NCERT-XII, Pg. # 101

180) NCERT Pg. # 84